細菌性疾病 (II)

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細菌性疾病 (II)

* 豬丹毒
* 豬沙門氏菌症
* 滲出性表皮炎
* 布氏桿菌症

豬丹毒 (Swine Erysipelas)

關聯 (Relevance)

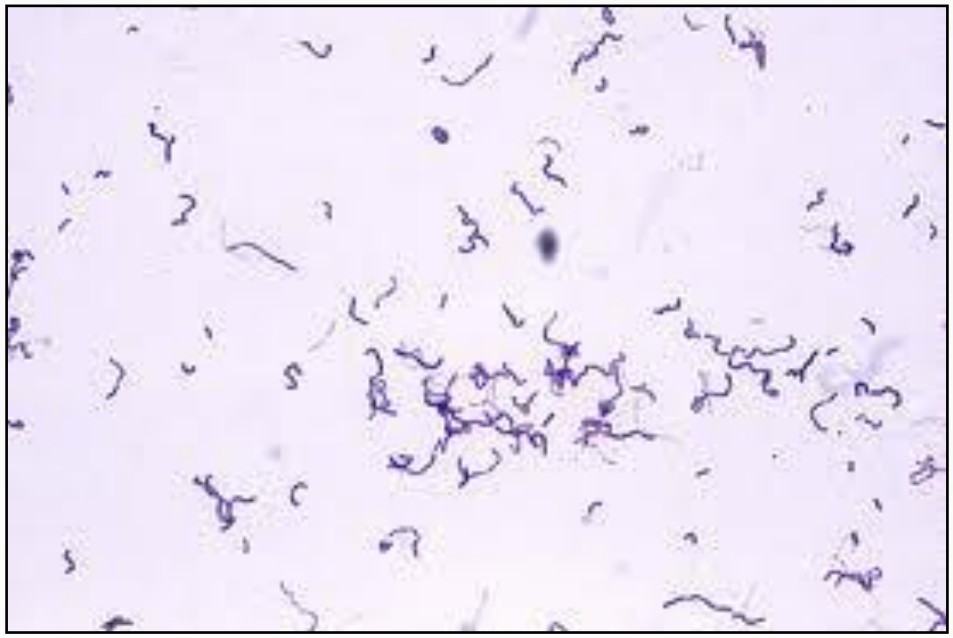
Swine erysipelas, when uncontrolled, is economically significant and capable of affecting all stages of pork production.

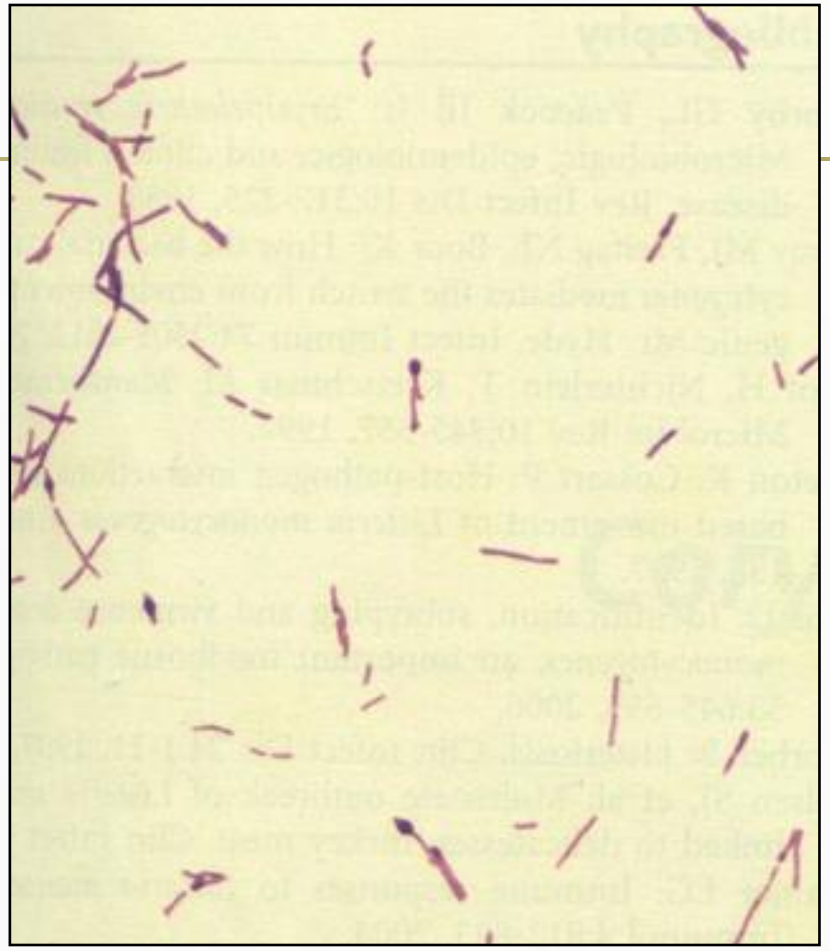
The greatest losses can be attributed to cases of sudden death and acute septicemia in grow–finish pigs.

A frequent sequel of surviving an acute infection is chronic lameness and arthritis, which results in poor growth and abattoir condemnations.

病因 (Etiology)

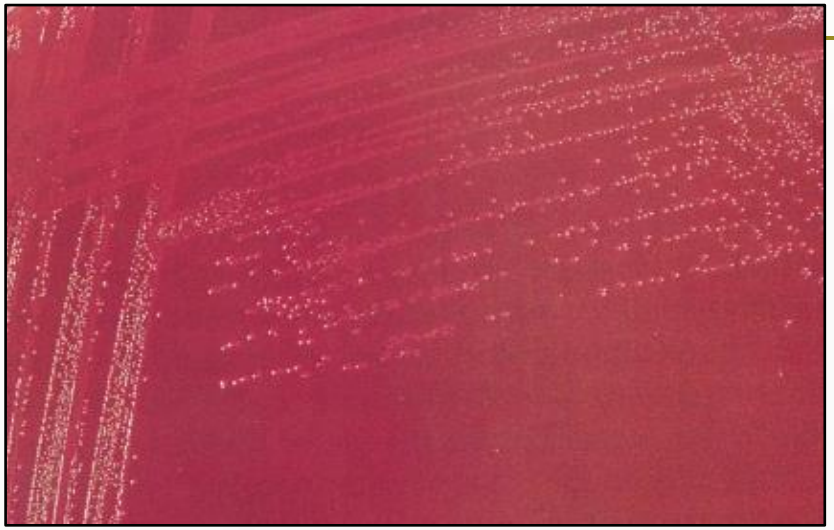
* *Erysipelothrix rhusiopathiae* (紅斑丹毒絲狀菌)
* Gram-positive, slender, non-spore-forming, nonmotile rods; facultatively anaerobic, slight bent bacillus, tend to form elongated filaments and sometimes pleomorphic



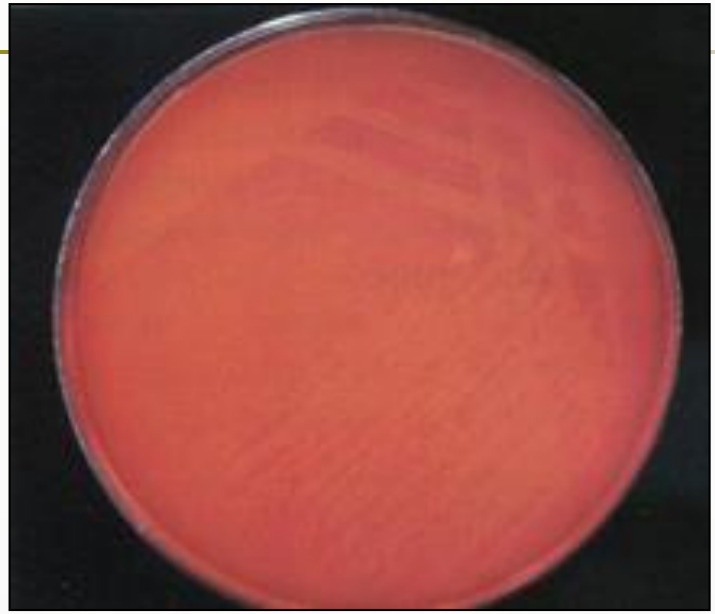


Erysipelothrix rhusiopathiae decolorize readily and may appear Gram-negative

Murray et al., *Medical Microbiology*, Elsevier, 2009



E. rhusiopathiae showing non-hemolytic colonies after 24 hours' incubation on sheep blood agar



E. rhusiopathiae colonies, showing hemolysis after 48 hours' incubation on sheep blood agar

公共衛生 (Public health)

* *E. rhusiopathiae* is zoonotic.

* Most cases occur via scratches or puncture wounds of skin as an occupational disease in those whose jobs are closely related to infected animals or their products.

* Localized infections, called erysipeloid (類丹毒, acute localized painful cellulitis with reddening of the skin), occur occasionally in people who work with infected animals or process meats, poultry, fish or animal by-products.



Erysipeloid. Classic purplish-red ring lesions caused by *Erysipelothrix rhusiopathiae*. (Courtesy of Bruce W. Halstead, MD, Colton, CA.)

Noonburg. *J Am Acad Orthop Surg 2005; 13:243-253.*

公共衛生 (Public health)

* Therefore, in humans, it is an occupational disease of butchers, meat processors, farmers, poultry workers, fish handlers, and veterinarians.

* Erysipeloid has been known by such names as:

  whale finger

  fish handler's disease

  pork finger

流行病學 (Epidemiology)

- *E. rhusiopathiae* is a ubiquitous microorganism that is distributed worldwide.

- It can be recovered on the tonsils or in the digestive tract of many domestic and wild animals.

- Colonization is particularly high in swine and turkeys.

- *E. rhusiopathiae* is spread among swine by carriers in the herd. Carriers disseminate the microorganisms in their feces and oronasal secretions. Carrier pigs are an important source of infection.

流行病學 (Epidemiology)

- Soil rich in organic matter or groundwater contaminated with animal waste can facilitate animal-to-animal spread.

- Fish meal fed to swine as a protein supplement sometimes contains *E. rhusiopathiae* once present in fish slime.

致病機制 (Pathogenesis)

* Ingestion of contaminated feed and water usually allows *E. rhusiopathiae* to gain access to the body, probably through the tonsils or other lymphoid tissue of the digestive tract.

* Natural infection can result from infected skin wounds.

* *E. rhusiopathiae* produces neuraminidase, an enzyme which may enhance cell penetration. Neuraminidase has been proposed as a virulence-associated factor and may contribute to tissue invasion and vascular injury.

致病機制 (Pathogenesis)

- Vascular damage leads to thrombosis and interference with microcirculation in capillaries and venulus. The capsule can resist phagocytosis.

- In chronic arthritis cases, there is vasculitis with exudation of fibrin into perivascular tissues and into the joints. There is also a marked villous proliferation of the synovial membrane (滑液膜) and joint lesions are more proliferative than exudative. Pannus (血管翳, abnormal layer of fibrovascular tissue) may develop and eventually result in destruction of articular cartilage and ankylosis (關節黏連) of the affected bones.

致病機制 (Pathogenesis)

- Valvular endocarditis is less common than joint involvement. Emboil of *E. rhusiopathiae* are believed to cause inflammation of blood vessels within heart valves.
- SpaA-surface protective antigen A: adherence
- RspA, RspB: surface proteins involving in biofilm formation

臨床症狀 (Clinical Signs)

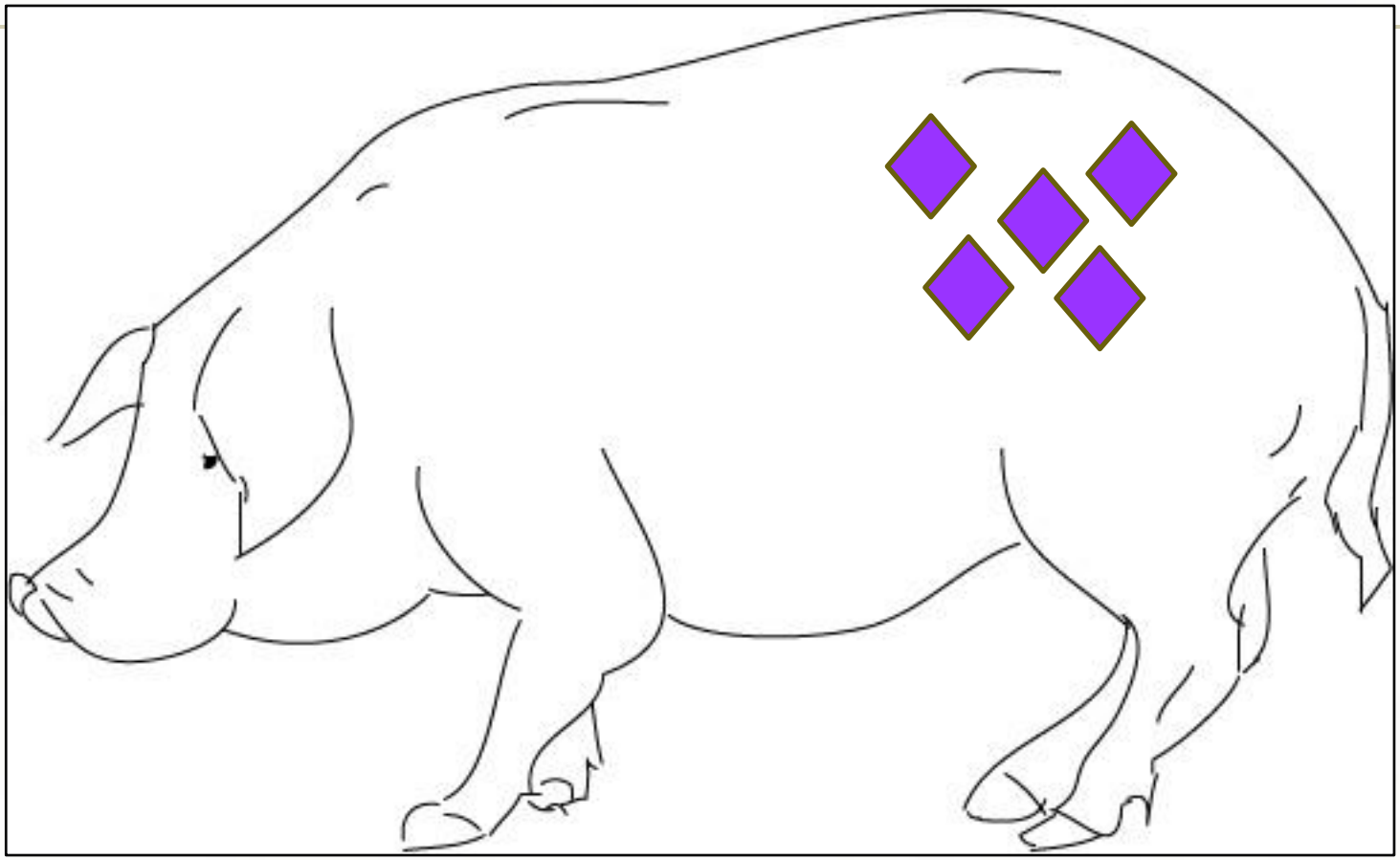
Acute

- Acute outbreak of SE occur in large hogs, approaching market weight. Acutely infected pigs are febrile (40-42°C).

- Sick pigs often have large areas of reddened or cyanotic skin, especially around the ear, snout, jowls, throat and abdomen.

- Some animals have discrete, raised, red-to-purple areas of skin with a rhomboid or diamond shape 2-3 days after exposure to the pathogen, usually more obvious on white pigs. On dark skinned pigs, those lesions may be visible only as slightly raised patches of hair.

皮膚紫紅色菱形或四角形突起斑疹







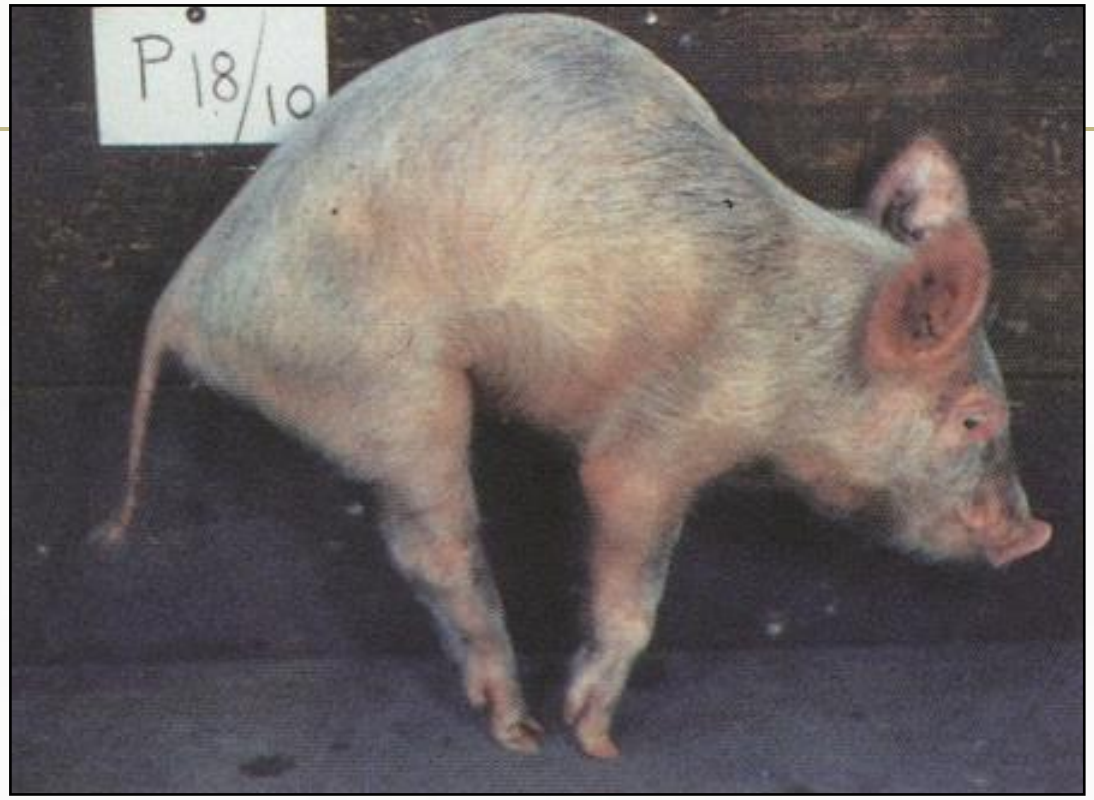
急性豬丹毒感染，皮膚紫紅色菱形或四角形突起斑疹，一 般稱之diamond-skin病變。指壓後會褪色，痊癒後壞死脫落。

劉振軒等著 豬病與屠宰衛生檢查圖譜 **1997** 台灣養豬科學研究所

臨床症狀 (Clinical Signs)

- The leg joints of sick pigs are painful and may or may not be swollen. They resist getting on their feet. When forced to arise, they often have to be assisted. They squeal with pain and soon lie down again.

- While standing, the feet are carried well under them and the head is hung dejectedly, giving the back line a marked arched appearance.



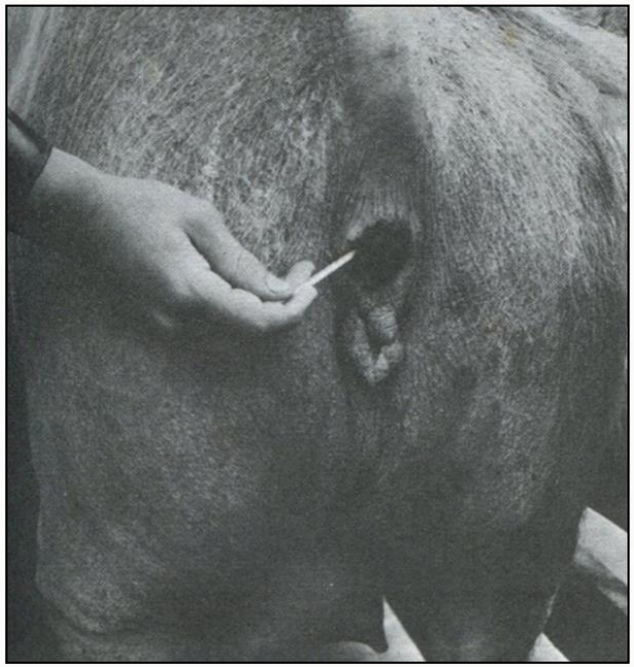
Affected animals are reluctant to get to their feet or take weight on their legs. They often stand with all four legs held together under their bodies.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.

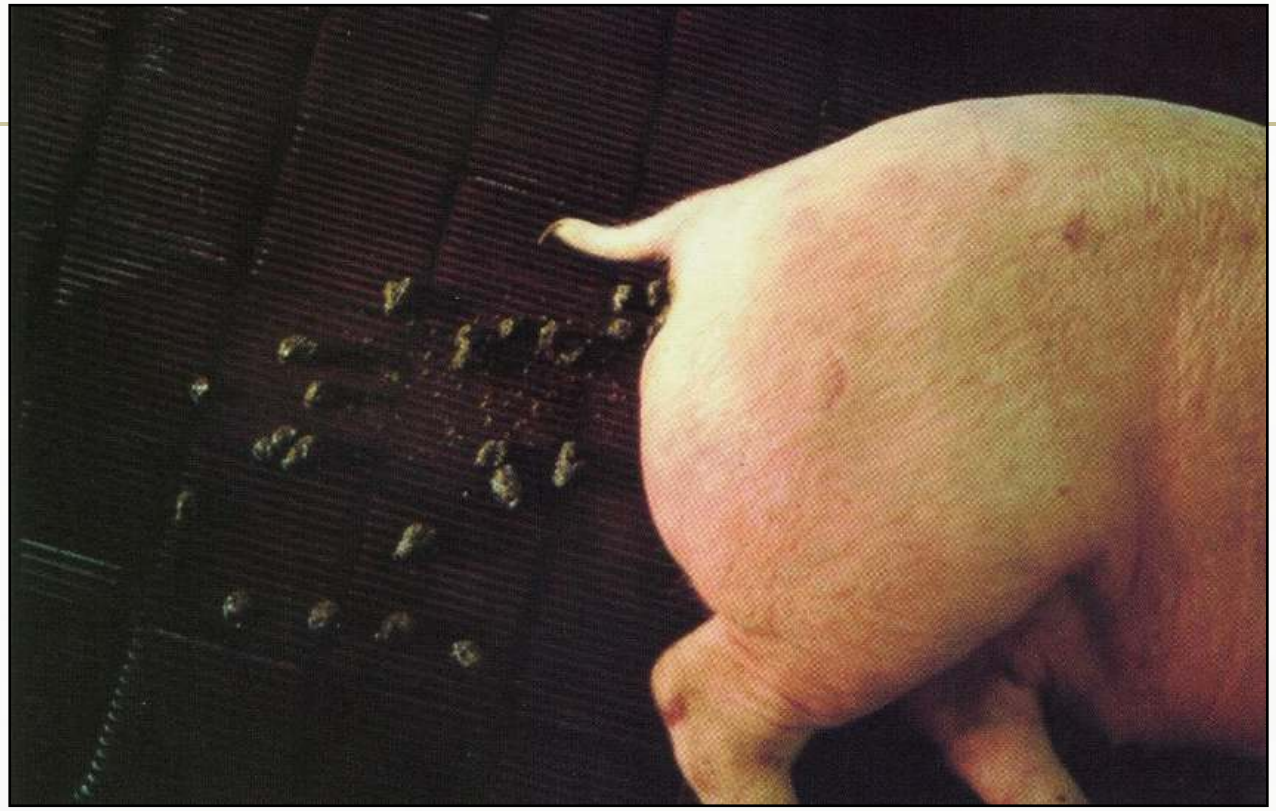
臨床症狀 (Clinical Signs)

* Most affected pigs will show partial or complete inappetence.

* Bowel movements are usually retarded and the feces firm and dry in pigs of market age and order.



TV VET book for pig farmers, 1967, Farming Press Ltd



病豬體溫升高可達40-41ºC，常見病豬排出如羊隻糞便呈顆粒狀

實用常見豬病防治圖說　2007

臨床症狀 (Clinical Signs)

Subacute

Milder systemic illness with cutaneous lesions.

Chronic

arthritis and/or valvular endocarditis; chronic skin necrosis may occur.

病變 (lesions)

* At necropsy of pigs died of acute erysipelas, there may or may not be discrete skin lesions.

* Lymph nodes are congested, often quite edematous.

* Petechial and ecchymotic hemorrhages usually are present on the cortex of the kidneys.[腎皮質常可見點狀及瘀斑性出血]

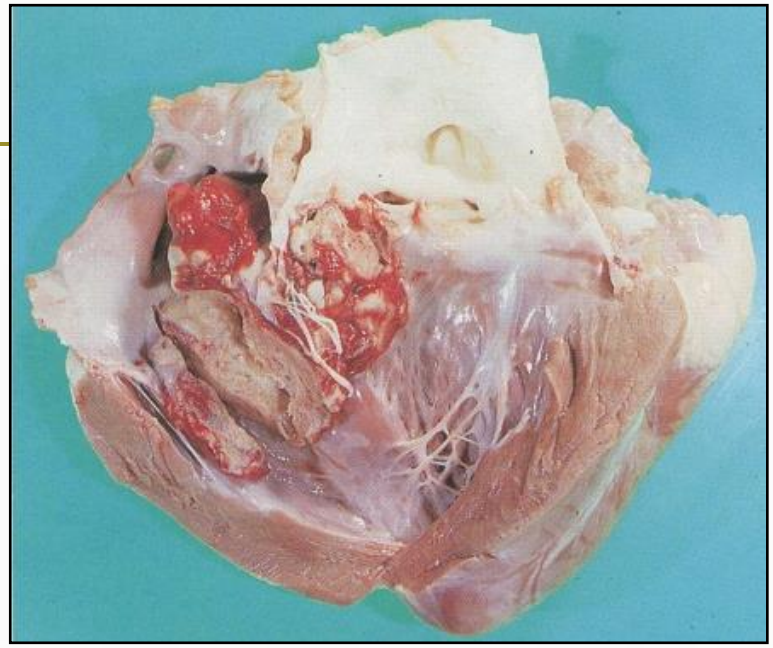
* The predominant lesion of chronic SE is a proliferative, nonsuppurative arthritis, occurring most commonly in hock, stifle, elbow, and carpal joints with occasional spondylitis.

病變 (lesions)

* Vegetative endocarditis may generate septic emboli, resulting in infarction of the kidney, spleen, and other organs.



The carcass of this pig, which died from vegetative endocarditis, shows this congestion over the hindquarters and on the ears.



A heart from a pig with chronic erysipelas vegetative endocarditis. Note the large vegetation on the atrio-ventricular valves.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.

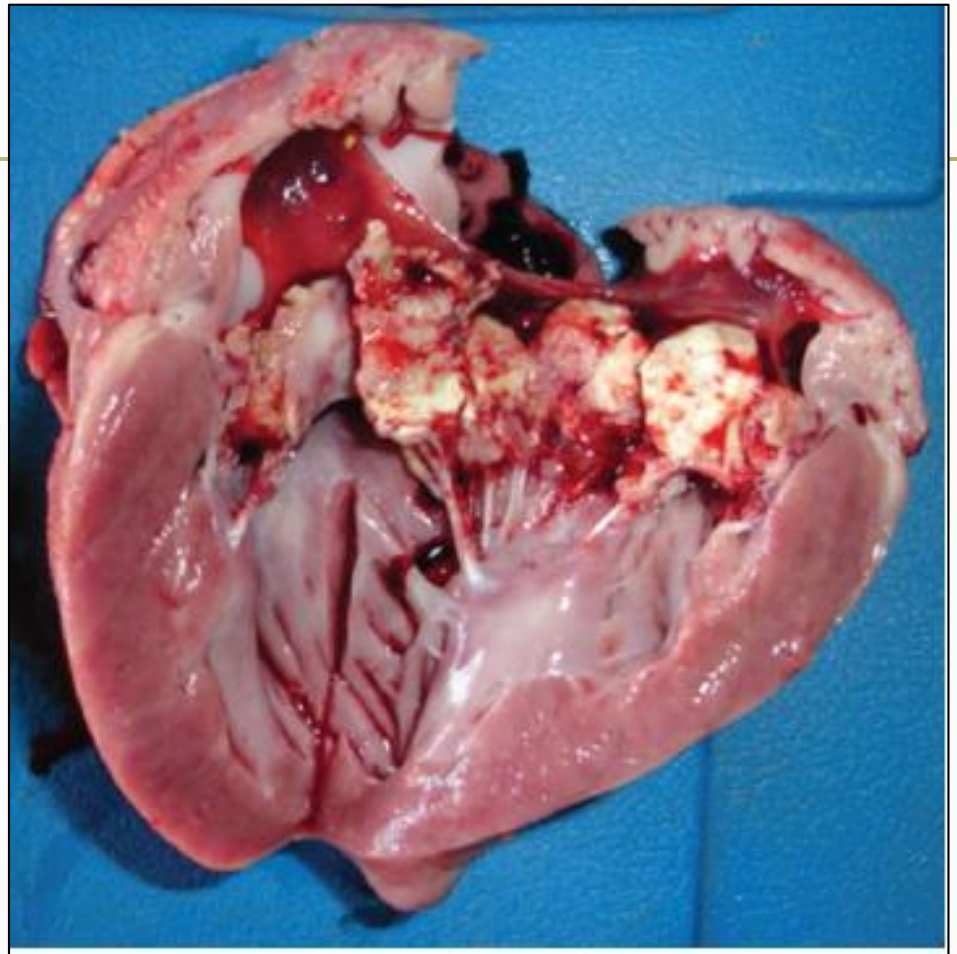
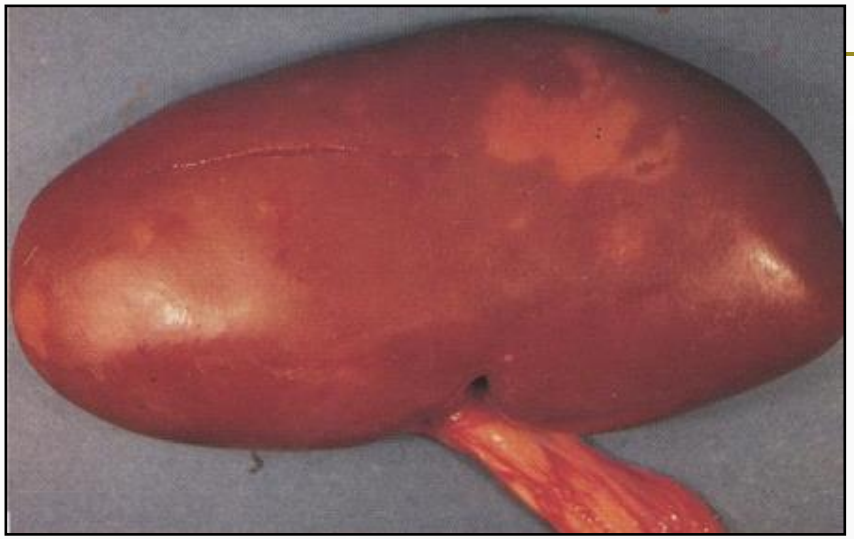


Figure 53.4 Valvular endocarditis in a pig with naturally occurring swine erysipelas. Note the proliferative granular growth on the heart valve. *Source:* Courtesy of Dr. João Gomes-Neto.



A kidney from a pig with vegetative endocarditis resulting from chronic erysipelas. Note the pale area of the infarct on the cortex of the kidney at top right.



A kidney from a pig with chronic erysipelas vegetative endocarditis, sectioned to show a pale infarct caused by an embolus.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



豬丹毒所引起之腎臟瀰漫性點狀出血

劉振軒等著 豬病與屠宰衛生檢查圖譜
1997 台灣養豬科學研究所

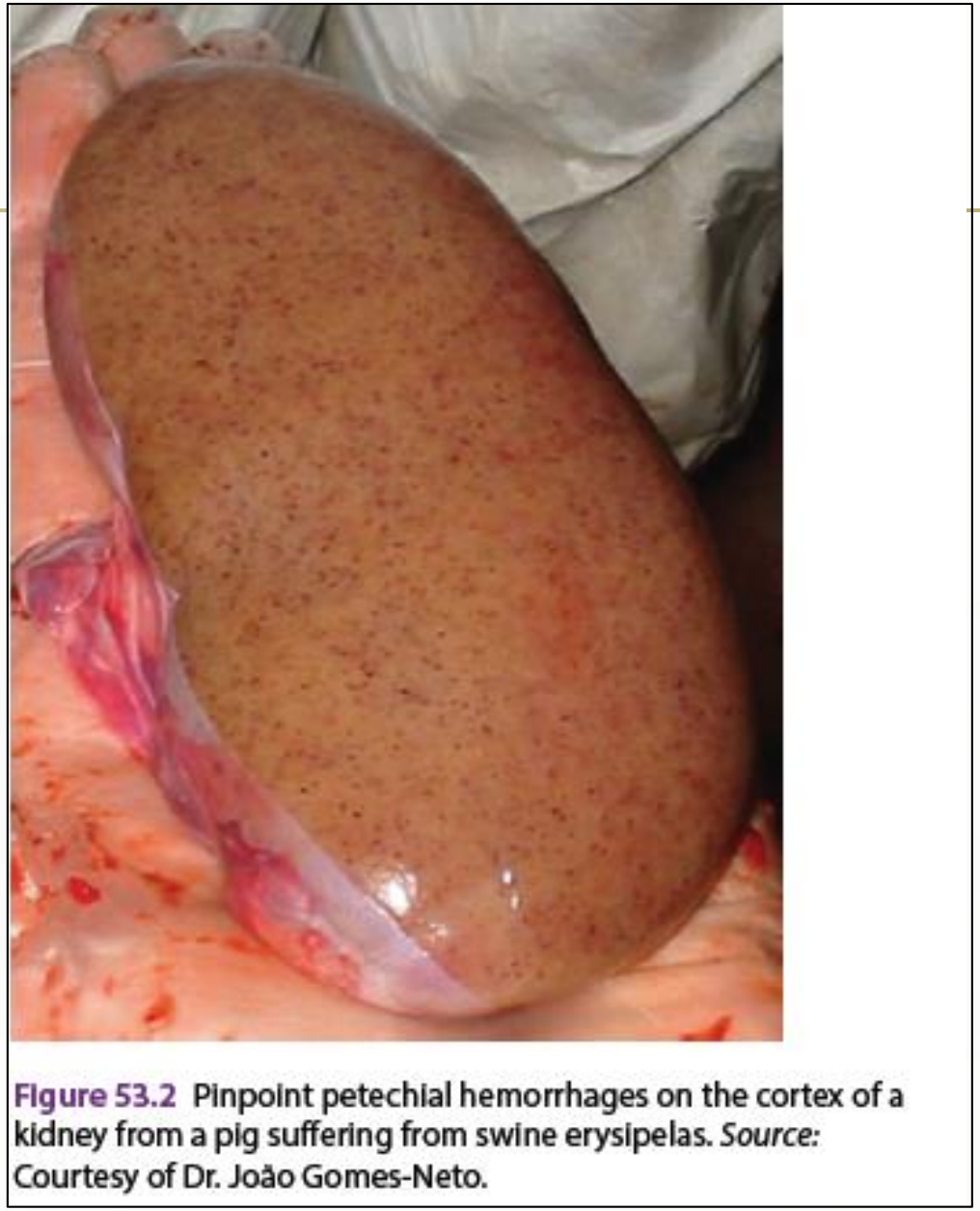
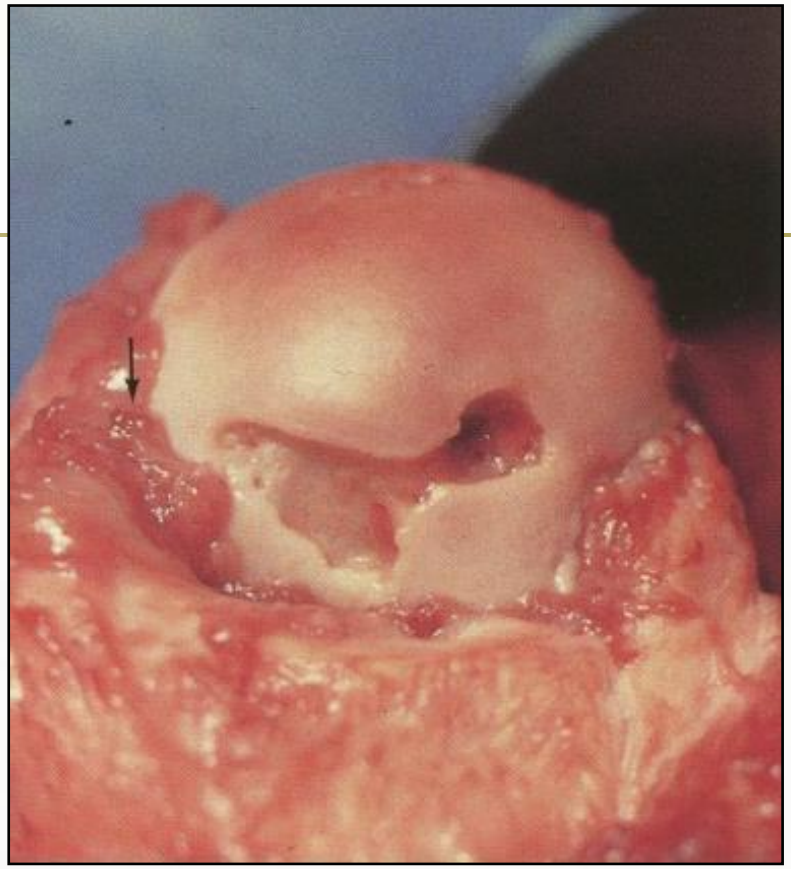


Figure 53.2 Pinpoint petechial hemorrhages on the cortex of a kidney from a pig suffering from swine erysipelas. *Source:* Courtesy of Dr. João Gomes-Neto.



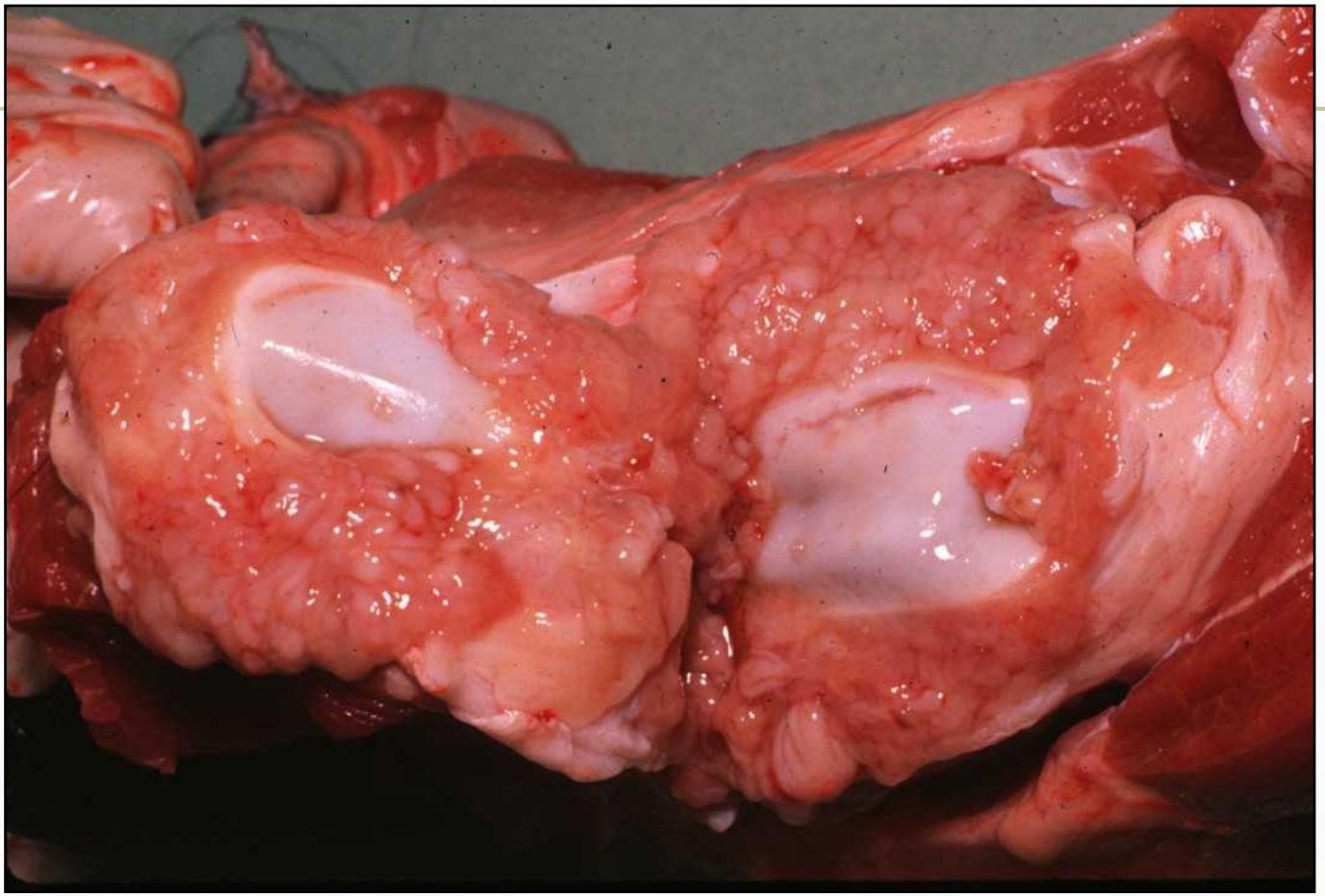
Spleen from a 70 kg pig which died with vegetative endocarditis. Infarcts can be seen as dark patches.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



Arthritis in chronic erysipelas from infection of the joint cavity and the subsequent proliferation of the synovial tissue. The femoral head shown here illustrates the hyperaemia and proliferation of the synovial membrane (arrow) and the erosion of the cartilage which occasionally accompanies it.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



Chronic severe proliferative synovitis-Erysipelas
(Courtesy of Dr. Madson, Iowa State University)

治療與控制 (Treatment and control)

- Treatment of choice: penicillin. Treatment is most effective when initiated early. Chronic arthritis and valvular endocarditis respond poorly to antimicrobial therapy.

- Prevention is best achieved by regular vaccination using killed bacterins or attenuated vaccines. The formalin-killed, aluminum-hydroxide-absorbed bacterin confers an immunity that protects growing pigs from acute disease until they reach market age.

- A combination of regular vaccination, along with good sanitation, the elimination of suspected carriers and appropriate isolation and quarantine measures usually will control swine erysipelas.

Example immunization program

* 經產母豬：每胎離乳後配種前補強注射一次
* 更新種公及女豬：配種前6週完成一次免疫
* 種公豬的免疫：每半年補強注射一次
* 保育豬：6週齡時注射一次

112年度國考試題

在豬場發現外觀如下圖之豬隻，下列有關此疾病之敘述何者正確？

A.主要藉由配種傳播

B.致病原為人畜共通傳染病原

C.致病原為革蘭氏陰性桿菌

D.致病原為一螺旋體



112年度國考試題

下列那一種病原廣布於土壤、水、腐敗的有機質、甚至魚類體表黏液，是一種人畜共通傳染病原。此病原感染豬隻後，若造成慢性感染，常見增殖性、非化膿性關節炎？

A. *Salmonella* Typhimurium

B. *Brucella suis*

C. *Yersinia pestis*

D. *Erysipelothrix rhusiopathiae*

110年度國考試題

下列何種病會產生急性高熱的症狀？

A.髮癬菌症

B.豬丹毒

C.水腫病

D.副結核病

110年度國考試題

下列何者為治療豬丹毒的首選藥？

A. amikacin

B. penicillin

C. sulfamethazine

D. sulfamonomethoxin

109年度國考試題

下列有關豬丹毒之敘述，何者錯誤？

A. 慢性豬丹毒的症狀皮膚可見紫紅色菱形或四角形突起斑疹

B. 急性豬丹毒症狀有發燒現象

C. 致病原是一種人畜共通傳染病原

D. Penicillin是首選治療藥物

107年度國考試題

豬丹毒是由一種革蘭氏陽性細菌感染所造成，有關此細菌之敘述，下列何者正確？

A. 是一種球菌

B. 會感染火雞

C. 具有運動性

D. 氧化酶（oxidase）反應為陽性

106年度國考試題

下列有關*Erysipelothrix rhusiopathiae*之描述，何者錯誤？

A.具有運動性之革蘭氏陽性菌

B.屬於人畜共通傳染性病原

C.健康豬隻扁桃腺可能帶有此病原

D.感染豬隻可由糞便或尿液排出此細菌

106年度國考試題

下列有關豬丹毒桿菌之敘述，何者正確？

A.可分泌腸毒素

B.可在加工肉品中存活數月

C.只感染豬隻

D.無特效藥，可用高免疫血清治療

106年度國考試題

豬隻罹患下列何種疾病於急性時會猝死，但轉為慢性時則有跛行或瓣膜栓子脫落致死？

A.豬丹毒

B.豬第二型環狀病毒感染

C.豬腎蟲症

D.豬小病毒感染

105年度國考試題

下列何種豬隻疾病在急性期皮膚可能會呈現紅紫色菱形突起之病變？

A.豬丹毒

B.萎縮性鼻炎

C.沙門氏菌症

D.滲出性表皮炎

104年度國考試題

下列有關豬丹毒（swine erysipelas）的敘述，何者正確？

A. 病原為一種病毒

B. 好發於哺乳仔豬

C. 主要經由呼吸道感染

D. 病原也可感染人

104年度國考試題

下列有關豬丹毒（swine erysipelas）之描述，何者錯誤？

A. 可引起腎臟皮質部出血點

B. 可引起瓣膜性心內膜炎

C. 急性型可造成皮膚菱形斑病變

D. 急性型可造成關節炎

102年度國考試題

下列有關豬丹毒桿菌之敘述，何者正確？

A. 主要藉由配種傳染

B. 屬於厭氧菌

C. 屬於革蘭氏陽性菌

D. 會分泌毒素

101年度國考試題

豬丹毒之病原為下列何者？

A. *Erysipelothrix rhusiopathiae*

B. *Haemophilus suis*

C. *Haemophilus parasuis*

D. *Streptococcus suis*

100年度國考試題

豬皮膚有菱形紅斑之病變時，可懷疑罹患下列何種疾病？

A. 豬生殖與呼吸綜合症

B. 豬沙門氏菌症

C. 破傷風

D. 豬丹毒

97年度國考試題

下列有關豬丹毒的敘述，何者正確？

A. 無菌苗可預防

B. 感染豬的腎臟出現壞死病灶

C. 好發於哺乳仔豬

D. 可以用抗生素治療

97年度國考試題

下列有關豬丹毒之敘述，何者正確？

A.由病毒引起

B.常好發於肥育豬

C.病原菌是巴氏桿菌

D.非人畜共通傳染病

97年度國考試題

下列有關豬丹毒之敘述，何者錯誤？

A.急性感染會發高燒

B.台灣已有病例發生

C.可造成心內膜炎

D.鏈黴素是首選抗生素

96年度國考試題

下列有關豬丹毒之敘述，何者正確？

A. 病原可分泌毒素

B. 可經由消化道和創傷感染

C. 病原是梭菌

D. 目前尚無疫苗可使用

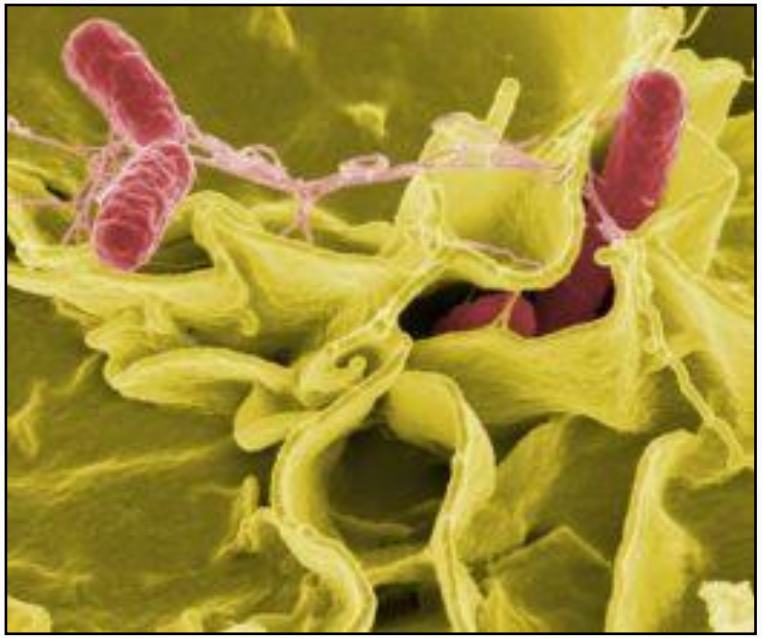
豬沙門氏菌症 (swine salmonellosis)

關聯 (Relevance)

Salmonella infections of swine are of concern for two major reasons:
The first is clinical disease in pigs (salmonellosis), and the second is that swine can be infected with a broad range of *Salmonella* serotypes that can potentially contaminate pork products and pose a threat to human health.

病因 (Etiology)

* *Salmonella*: Gram-negative, facultatively anaerobic, non-lactose-fermenting bacilli; facultative intracellular pathogens.



- *Salmonella*屬可分為*Salmonella enterica*及*Salmonella bongori* 2個種

- *S. enterica*: 主要的病原 *S. bongori*: 多見於冷血動物

- 根據 White–Kauffmann–Le Minor 分類系統 (沙門氏菌血清分類系統)，*Salmonella* 已知約有 2,600以上種血清型 (serotypes)。

- *Salmonella enterica* serotype Typhimurium=*Salmonella* Typhimurium (Typhimurium是血清型)

與人類/動物相關的常見*Salmonella*血清型

serotype	host	note
Typhimurium	pigs, cattle, poultry, human	broad host range, nontyphoidal
Enteritidis	eggs, poultry, human	food poisoning
Choleraesuis	pigs, human	septicemia in pigs
Dublin	cattle, human	systemic infection in human
Gallinarum/Pullorum	poultry	fowl typhoid/pullorum disease
Typhi	human	typhoid fever
Paratyphi A, B, C	human	paratyphoid fever

病因 (Etiology)

The following two serotypes are associated with diseased swine:

serotype	Host association	Major disease in pigs
Typhimurium	broad host range	enterocolitis
Choleraesuis	pig-adapted	septicemia/systemic disease

公共衛生 (Public health)

- Nontyphoid salmonellosis (NTS) is one of the leading causes of foodborne illness in many countries.
- Less common than contaminated food, direct contact with clinically or subclinically infected animals can also be a source of human infection.
- Typhimurium, Heidelberg, Agona, Infantis are 4 serotypes commonly isolated from swine and are among the top 20 serotypes isolated from diseased humans.
- *S*. Choleraesuis is an important cause of invasive salmonellosis in humans in parts of Asia.

流行病學 (Epidemiology)

- 帶菌豬隻(carrier pig)為最重要的感染源。
- Concurrent infections and other stressors may precipitate clinical disease.

流行病學 (Epidemiology)

- Infection of swine with persistent shedding by one or more serotypes is common, but primary clinical disease caused by serotype other than Choleraesuis or Typhimurium is uncommon.

- Clinically normal swine may be infected and shed a variety of serotypes that pose little threat to pigs, but may pose risk for human through direct exposure or contamination of pork products.

- 任何*Salmonella* serotype對人類而言，都是病原菌!

Salmonellae in pork

- Infected pigs leaving the farms are considered the original source of abattoir contamination.

- The stress of transport and feed deprivation increases shedding from unapparent carriers, which then contaminate the environment of the truck and abattoir.

Salmonellosis in swine

* Most salmonellosis outbreak occur in intensively reared weaned pigs.

* Host-adapted *S*. Choleraesuis is isolated almost exclusively from diseased swine and is usually manifested as septicemia. Severe *S*. Choleraesuis infection occurs typically at around 12 to 14 weeks.

* Clinically ill pigs infected with *S*. Typhimurium develop enterocolitis and exhibit diarrhea and dehydration.

致病機制 (Pathogenesis)

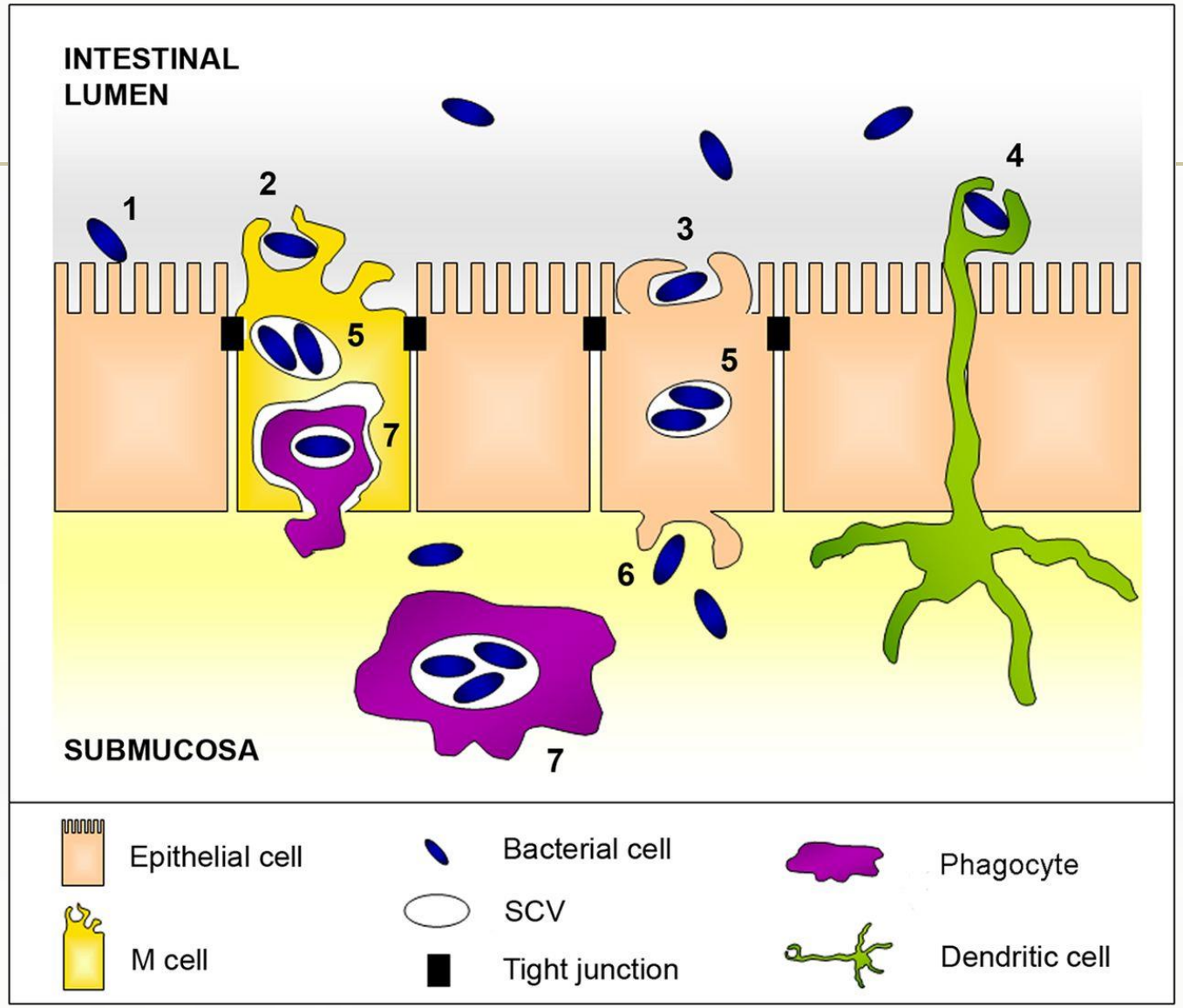
- Adherence
- Pathogenicity island (PAI)

  PAI是一段位於細菌染色體或質體上的基因群集，這些基因與致病能力有關。

  SPI-1: type III secretion system → epithelial invasion and membrane ruffling

  SPI-2: type III secretion system → intracellular survival and systemic dissemination
- Virulence plasmid

  此毒力質體帶有*spv* 基因組，使感染能從腸道局部擴散成全身性敗血症
- Acid tolerance and other stress-response mechanisms facilitate survival through the gastrointestinal tract.



Clin. Microbiol. Rev. 2013, 26: 308-341

臨床症狀與病變 (Clinical Signs and lesions)

- Septicemic salmonellosis

Clinical signs:

inappetent, lethargic, febrile (40.5-41.6 C), reluctant to move, huddled in the corner of a pen, or even dead, with cyanosis of extremities and abdomens.

Lesions:

lesions include cyanosis of ears, feet, tail, and ventral abdominal skin. Splenomegaly with less severe hepatomegaly, moist swollen gastrohepatic and mesenteric lymph nodes. Lungs are firm and resilient, diffusely congested, often with interlobular edema and hemorrhage. Miliary random white foci of necrosis in the liver.

臨床症狀與病變 (Clinical Signs and lesions)

- *Salmonella* enterocolitis

Clinical signs:

watery yellow diarrhea, initial without blood or mucus, febrile, decreased feed intake, dehydrated. Mortality usually is low.

Lesions:

Focal or diffuse necrotic enteritis, colitis, or typhlitis (盲腸炎).

沙氏桿菌之剖檢病變

* 敗血型沙氏桿菌症（豬隻主要死於敗血症）
  - 耳翼及四肢末端發紺
  - 胃黏膜鬱血繼之梗塞、出血性潰瘍
  - 脾腫大及鬱血，脾梗塞
  - 全身淋巴結壞死、出血、腫大（尤其腸繫淋巴結/耐過豬明顯）
  - 肺全面輕度堅實、鬱血、小葉間水腫、出血
  - 肝腫大、多發局部壞死；腎針點狀出血點
  - 耐過豬以壞死性結腸炎為主徵
* 腸炎型沙氏桿菌症（豬隻主要死於長期下痢）
  - 局部至瀰漫性壞死性結腸、盲腸炎，偶有小腸病灶
  - 腸繫淋巴結（尤其中後段腸管）極度腫大、壞死

如何察覺豬場中發生之沙氏桿菌症

- 豬群行為
  - 豬群交疊、沈鬱、死亡率提昇
- 外觀及臨床症狀
  - 敗血型：急速死亡、發紺、全身皮膚出血斑點
  - 腸炎型：6週齡左右開始長期而反覆性下痢
- 肉眼病變
  - 敗血型：全身臟器出血點、壞死點、梗塞 肝脾腫脹
  - 腸炎型：壞死性、潰瘍性腸炎



Septicemic salmonellosis

劉振軒等著 豬病與屠宰衛生檢查圖譜 1997
台灣養豬科學研究所



不願走動、群聚取暖



賴秀穗 臺大獸醫系



Piglet acute salmonellosis. Note the cyanosis of the ear tips.



Piglet acute salmonellosis. Sloughing of the ear tips may occur in some recovered cases.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



Cyanosis of ears, nose and extremities



Diarrhea is golden colored



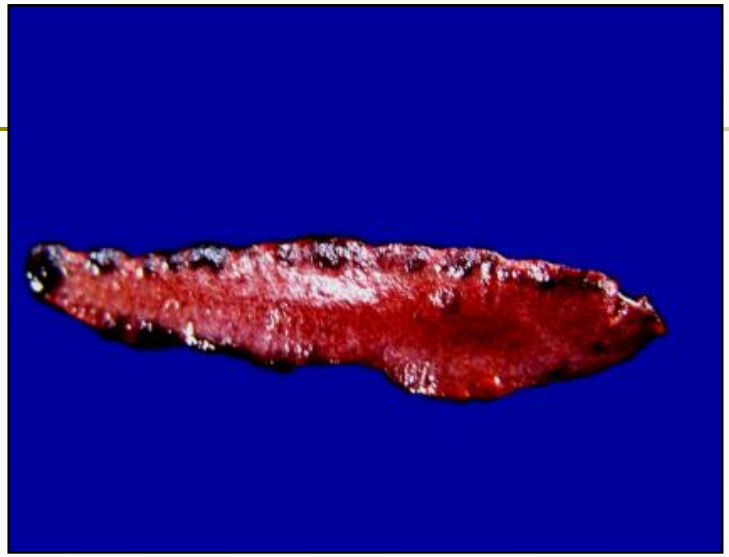
Enterocolitis

Dr. J. Carr



S. Choleraesuis感染，耳翼出現明顯cyanosis現象。(養豬業者常稱此病為: 紅耳朵)

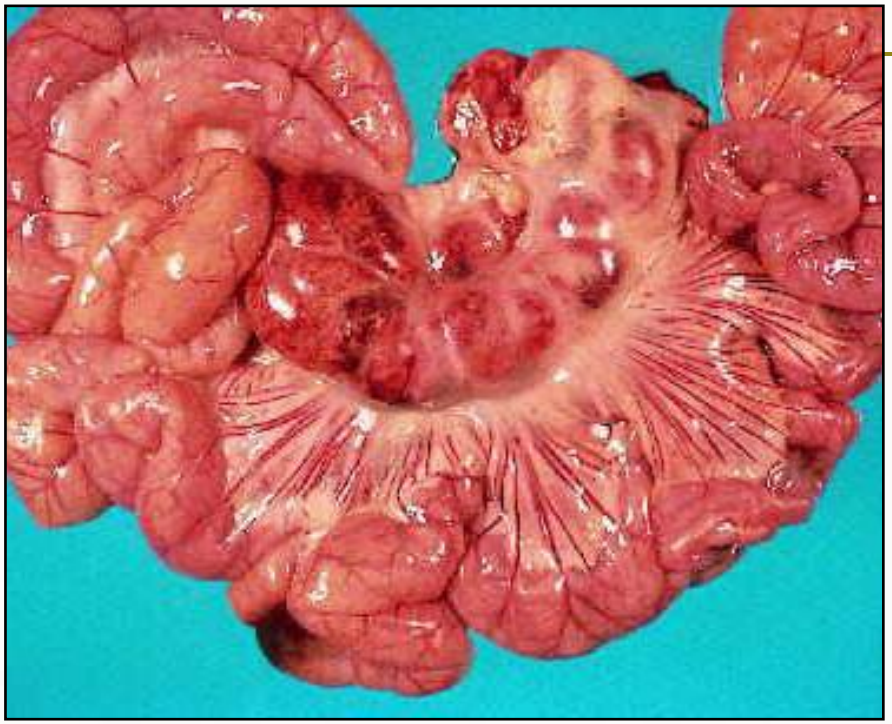
劉振軒等著 豬病與屠宰衛生檢查圖譜 **1997** 台灣養豬科學研究所



脾臟：周邊性梗塞



脾臟：極度腫大



Enlarged, hemorrhagic Lymph nodes

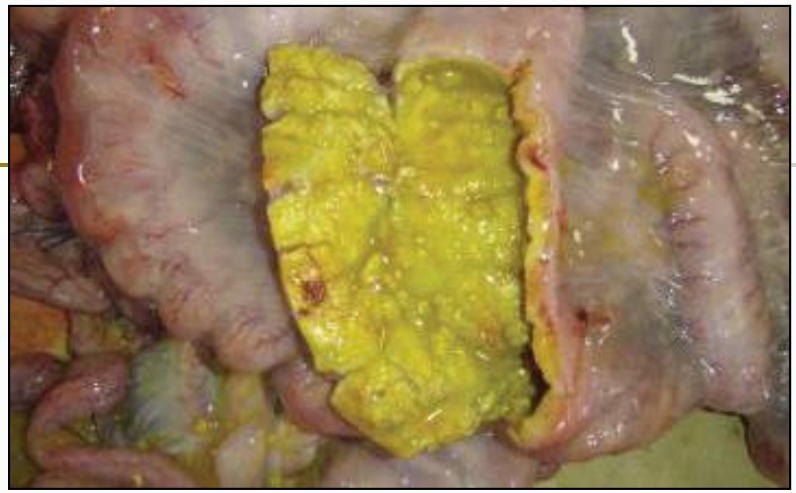


Markedly enlarged spleen

Courtesy of Dr. Madson, Iowa State University



Swollen lymph node



Necrotic enteritis

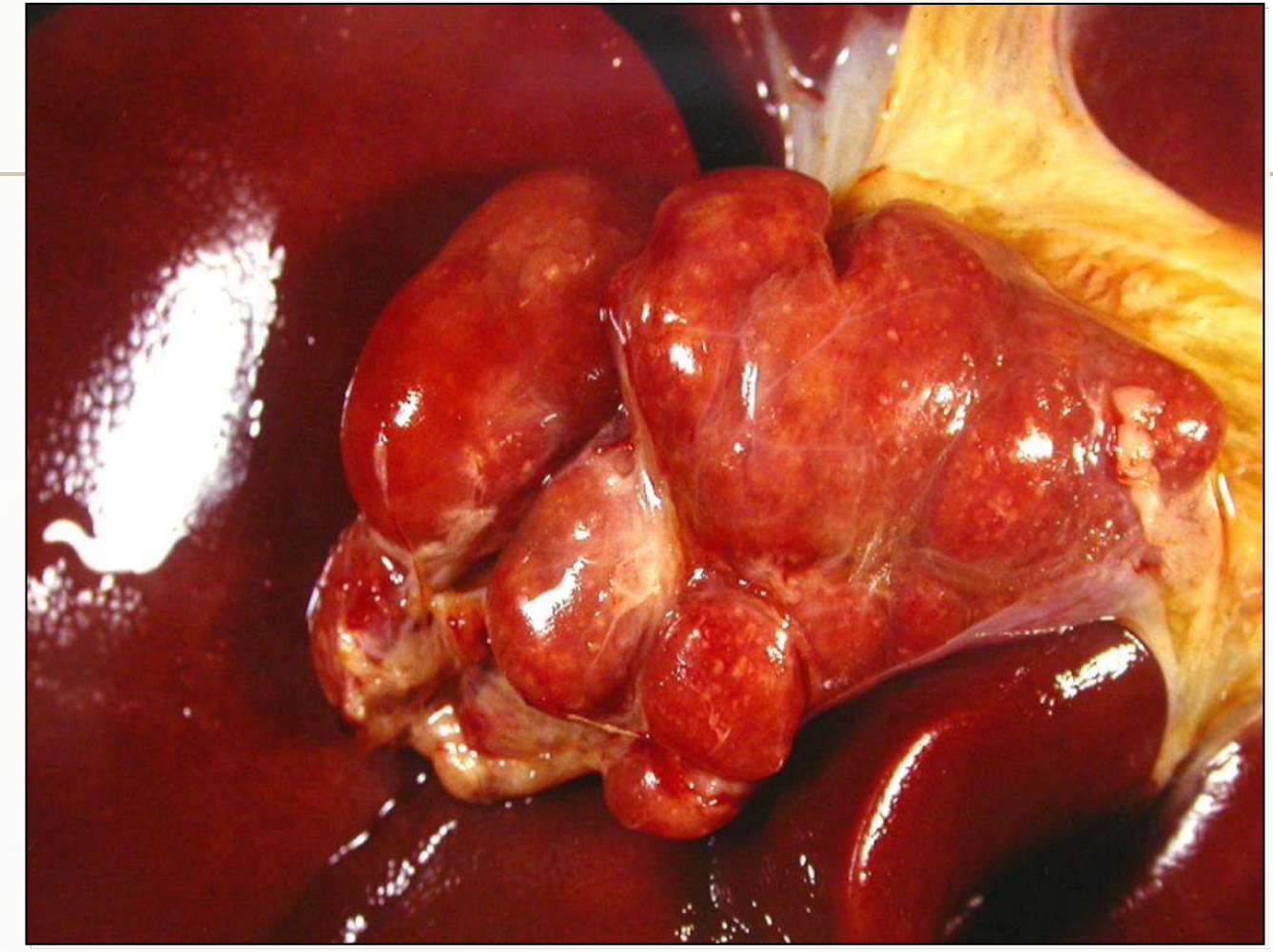


Lung blotchy consolidation





壞死性肝炎：多發局部針點狀壞死點，偶而可見出血點



肝門淋巴結(壞死性淋巴線炎) 腫大、多發局部點狀至斑狀壞死

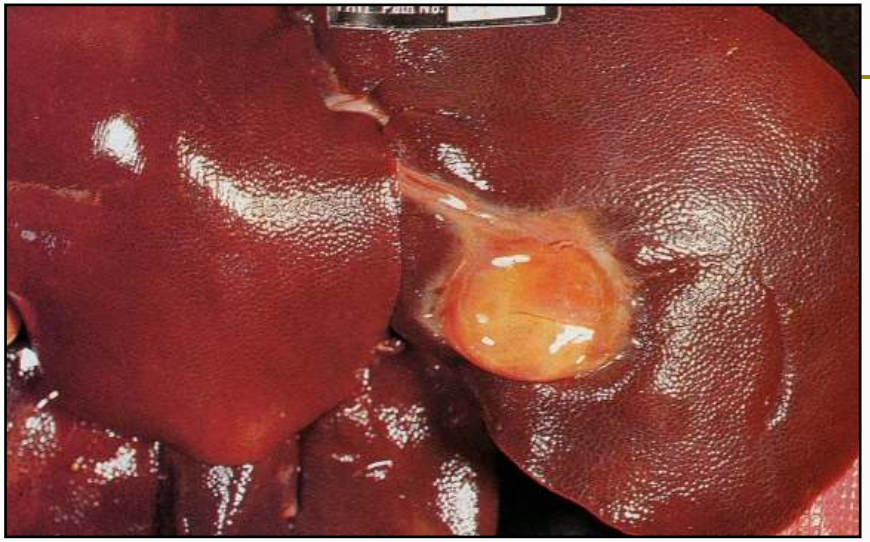


淋巴結切面：多發局部點狀至斑狀壞死



肺臟全面性水腫而使肺腫大，肺臟小葉間隔間隙淋巴管擴張，肺泡壁增厚，觸摸呈現實質化。

劉振軒等著 豬病與屠宰衛生檢查圖譜 1997 台灣養豬科學研究所



肝臟具點狀壞死之壞死性肝炎膽囊壁呈現水腫。



肝臟具黃白色點狀壞死特徵為凝固性壞死於病變區存在副傷寒小節(paratyphoid nodule)

Paratyphoid nodules are small foci of hepatic necrosis with inflammatory cell accumulation, classically associated with salmonellosis.

劉振軒等著 豬病與屠宰衛生檢查圖譜 1997 台灣養豬科學研究所



壞死性潰瘍性結腸炎：多發局部至瀰漫性壞死及潰瘍，病灶形成偽膜樣結構（慢性）



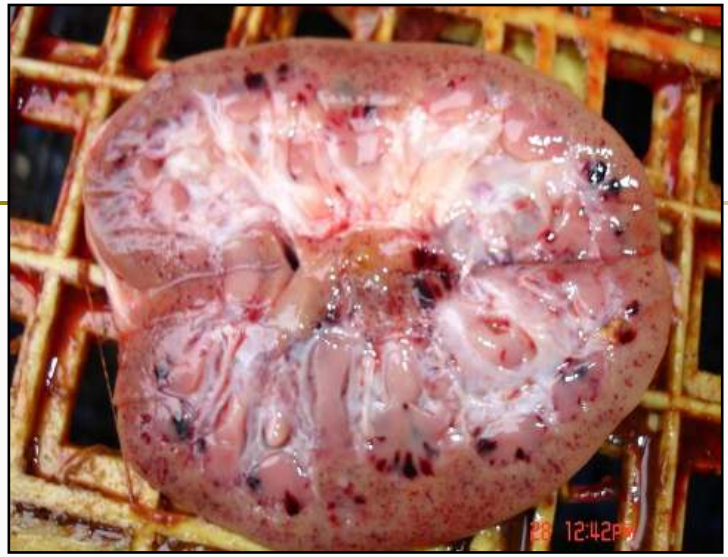
鈕扣狀潰瘍(button ulcers): 許多黃色如鈕扣狀的潰瘍，獨立散佈在結腸黏膜面為沙門氏桿菌腸炎的特徵。



除了鈕扣狀潰瘍外，左側結腸黏膜表面出現染成膽汁色的內容物。

劉振軒等著 豬病與屠宰衛生檢查圖譜 1997 台灣養豬科學研究所





SALMONELLA





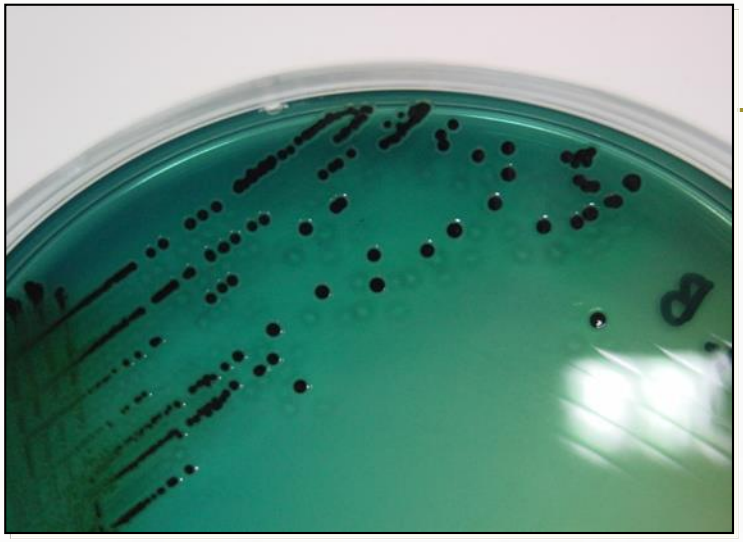
賴秀穗 臺大獸醫系

診斷 (Diagnosis)

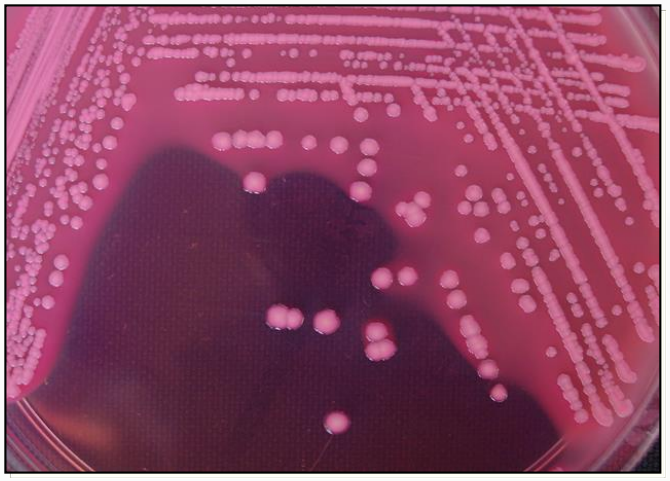
Bacterial isolation



Salmonella on MacConkey agar



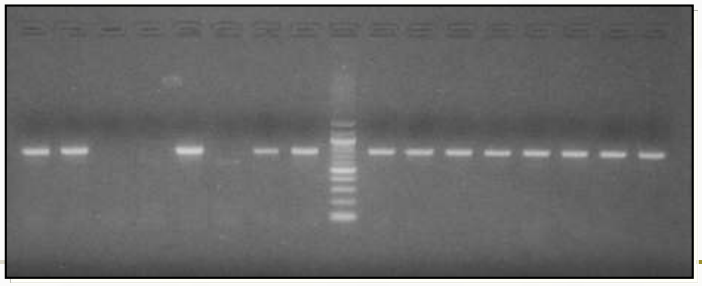
Salmonella on HE agar



Salmonella on SMID agar

診斷 (Diagnosis)

PCR

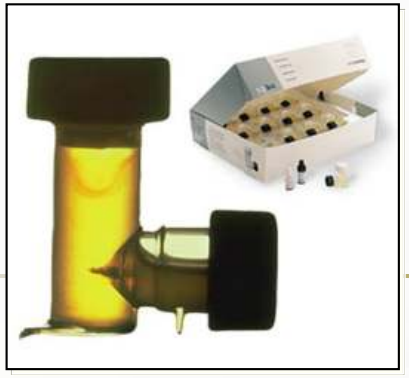


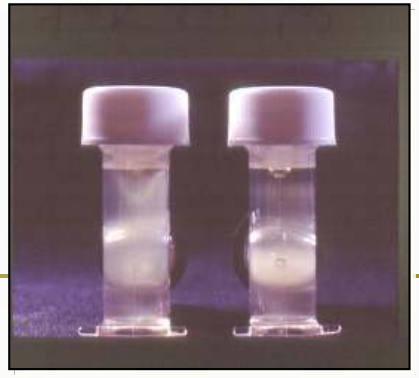


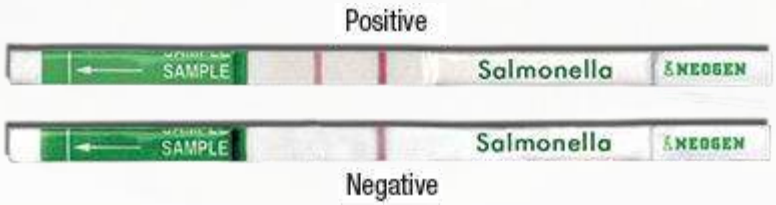


診斷 (Diagnosis)

Immunoassay









預防 (Prevention)

Complete exclusion of *Salmonella* from pig populations is difficult. Control focuses on reducing exposure, minimizing stress and shedding, improving hygiene and biosecurity, and vaccination where appropriate.

Killed vaccines generally induce less robust protection than live attenuated vaccines against intracellular *Salmonella* infection.

Some attenuated live vaccine demonstrated to have protective efficacy.

Examples of commercial vaccines



Enterisol *Salmonella* T/C®



Salmoporc

115年度國考試題

保育舍仔豬突發性下痢，病豬呈現消瘦及脫水現象，死亡豬隻解剖可見盲腸及結腸鈕扣樣潰瘍病變，最有可能爆發何種疾病？

A.球蟲症

B.豬流行性下痢

C.沙門氏菌症

D.豬赤痢

113年度國考試題

下列那一個選項與沙門氏菌感染豬隻所造成的症狀較無相關？

A. 敗血症

B. 滲出性皮膚炎

C. 腸炎

D. 肺炎

112年度國考試題

臨床上會造成豬隻呈現耳翼及四肢末端發紺等敗血症狀，最主要是由下列何種沙門氏菌血清型所致？

A. *Salmonella* Choleraesuis

B. *Salmonella* Heidelberg

C. *Salmonella* Derby

D. *Salmonella* Dublin

110年度國考試題

如下圖所示，此斃死豬全身皮膚可見塊狀紫斑及大小不一的出血點，剖檢觀察到脾臟腫大、肝臟腫大且有散發性的粟粒狀大的壞死白點。從病灶處分離出革蘭氏陰性桿菌，且在MacConkey agar上呈現無色菌落。其最有可能的致病原為下列何者？

A. *Escherichia coli*

B. *Salmonella Choleraesuis*

C. *Pasteurella multocida*

D. *Brucella suis*



109年度國考試題

一豬隻有耳翼及四肢末端發紺，剖檢可見脾臟腫大及淋巴結腫大壞死，結腸黏膜有許多黃色如鈕扣狀的潰瘍，肝臟也有黃白色點狀壞死。可從病灶分離出革蘭氏陰性桿菌，此病原最有可能為下列何者？

A. *Salmonella* Choleraesuis

B. *Bordetella bronchiseptica*

C. *Klebsiella pneumoniae*

D. *Pasteurella multocida*

109年度國考試題

離乳豬若感染一種革蘭氏陰性菌後，豬隻可能呈現的臨床症狀包括消瘦、下痢、生長遲緩等現象，其下痢便呈水樣黃色。下圖為剖檢後，切開病變的結腸，呈現其黏膜面。下列何者是最有可能的感染病原？

A. *Haemophilus parasuis*

B. *Lawsonia intracellularis*

C. *Salmonella* Typhimurium

D. *Escherichia coli*



108年度國考試題

下列有關豬隻沙門氏菌症之敘述，何者錯誤？

A. 常見的臨床症狀包含敗血症與腸炎

B. 病豬剖檢時，在肝臟有時可見黃白色點狀壞死病變

C. 會造成臨床症狀的沙門氏菌血清型以Choleraesuis與Typhimurium為主

D. 藉由配種傳播

106年度國考試題

Kaufmann-White 分類系統是應用於那一屬細菌的血清型分類？

A. *Escherichia*

B. *Klebsiella*

C. *Leptospira*

D. *Salmonella*

104年度國考試題

下列有關豬隻沙門氏菌症的敘述，下列何者正確?

A. 藉由空氣傳播

B. 為一週齡以下哺乳仔豬下痢的主因

C. 會造成臨床症狀的血清型以Pullorum為主

D. 可造成敗血症與肺炎

104年度國考試題

下列哪一種沙門氏菌血清型，相對於另外三個選項，有較廣泛的感染宿主?

A. *Salmonella* Arizonae

B. *Salmonella* Typhimurium

C. *Salmonella* Choleraesuis

D. *Salmonella* Gallinarum

103年度國考試題

下列何種分泌系統在沙門氏菌接觸細胞後，可將作用蛋白質（effector proteins）注射至細胞內，並誘導細胞的細胞膜起皺紋（membrane ruffling），而進入細胞內？

A. Type I

B. Type II

C. Type III

D. Type IV

102年度國考試題

下列何者病原於豬隻感染後，主要呈現敗血症，細菌培養在MacConkey agar上呈現無色的菌落？

A. *Streptococcus suis*

B. *Salmonella* Choleraesuis

C. *Escherichia coli*

D. *Brucella suis*

102年度國考試題

下列有關豬隻感染*Salmonella* Typhimurium及*Salmonella* Choleraesuis兩種沙門氏菌血清型的比較，何者正確？

A. *Salmonella* Choleraesuis在腸道黏膜造成的傷害比較嚴重

B. 豬感染*Salmonella* Typhimurium主要的症狀以肺炎為主

C. *Salmonella* Typhimurium感染動物種類比較多

D. 此二種血清型若培養在MacConkey agar上面，都會呈現粉紅色菌落

101年度國考試題

下列何者不是豬沙門氏菌症的病變之一？

A.結腸鈕扣狀潰瘍

B.淋巴結腫大充血

C.腦膜腦炎

D.腎臟出血

101年度國考試題

*Salmonella dublin*主要造成那種動物的敗血症？

A.犬

B.牛

C.馬

D.豬

101年度國考試題

下列有關豬隻腸炎型沙門氏桿菌症（salmonellosis）的敘述，那些正確？①病變侷限於大腸部位

②多發生在保育後期及肉豬前期　③病變多為纖維素壞死性炎症　④鈕扣狀潰瘍為其特徵性慢性病灶

A.①②③

B.①③④

C.②③④

D.①②④

100年度國考試題

下列何者無法藉由空氣傳播？

A. 黴漿菌性肺炎

B. 口蹄疫

C. 萎縮性鼻炎

D. 沙門氏菌症

100年度國考試題

亞急性及慢性豬隻沙門氏菌症會在下列哪個部位產生潰瘍病變?

A. 胃

B. 十二指腸

C. 結腸

D. 直腸

99年度國考試題

下列何種沙門氏菌，最易引起豬的肺炎？

A. *Salmonella derby*

B. *Salmonella choleraesuis*

C. *Salmonella paratyphi*

D. *Salmonella typhimurium*

97年度國考試題

下列有關沙門氏菌症之敘述，何者正確？

A. 可由節肢動物叮咬而傳染

B. 可引起敗血症和肺炎

C. 主要造成仔豬腦膜炎症狀

D. 是造成哺乳豬下痢的主要原因

96度國考試題

下列有關豬沙門氏菌症之敘述，何者錯誤？

A. 致病菌是革蘭氏陰性菌

B. 肝臟腫大，可能有小壞死點

C. 主要病菌是 *Salmonella choleraesuis*

D. 尚無疫苗可供預防

滲出性表皮炎(exudative epidermitis)
油脂豬病 (greasy pig disease)

關聯 (Relevance)

* exudative epidermitis, also known as greasy pig disease, occurs worldwide and is the most common staphylococcal skin disease in swine.

* A disease seen from 7 days to 10 weeks (sucking and newly weaned piglets).

* The disease is characterized by the accumulation of greasy exudate on lesion areas. Affected pigs develop generalized nonpruritic exudative dermatitis–epidermitis

病因 (Etiology)

- *Staphylococcus hyicus* (豬葡萄球菌)

  Gram-positive cocci, normal bacterial flora on the skin of adult swine. The organism can also be recovered from the environment in swine facilities.



Staphylococcus hyicus

Gram-positive coccus that forms 3-4 mm porcelain-white nonhemolytic colonies on sheep blood agar.

http://pictures.life.ku.dk/atlas/microatlas/veterinary/bacteria/Staphylococcus_hyicus/

公共衛生 (Public health)

- *Staphylococcus hyicus* is primarily an animal pathogen and is not considered an important zoonotic pathogen.

流行病學 (Epidemiology)

- Infection is carried by the sow or gilt on her skin or within vagina.
- Abrasion on the feet and legs or lacerations on the body precede infection. Such injuries are usually caused by fighting or abrasive surfaces such as new concrete.[skin abrasion / trauma precedes infection]
- One or two piglets or occasionally the whole litter may be affected.
- *S. hyicus* is very resistant to adverse conditions (as most staphylococci are) and can persist in the environment for a long period of time.

致病機制 (Pathogenesis)

- Virulence is close associated with the production of exfolitative toxins (表皮剝落毒素)
- The toxin separates the cells in the epidermis, allowing rapid intraepidermal spread of the bacteria.
- Exfoliation of the skin is accompanied by excess sebaceous secretion and serous exudate.
- These toxins are similar to the toxins produced by strains of *S. aureus* that are isolated from staphylococcal scaled skin syndrome (SSSS) in humans.

Scalded skin syndrome in human (人類表皮剝落症候群)





Murray et al: Medical Microbiology, 6th Edition.
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致病機制 (Pathogenesis)

* Several exfoliative toxins have been identified:

From Denmark:

ExhA

ExhB

ExhC

ExhD

From Japan:

ShetA

ShetB

臨床症狀 (Clinical signs)

* Piglets usually develop the disease between 4-6 days and 5-6 weeks of age.

* Clinical signs begin with listlessness or depression and anorexia, dejection (沮喪) and a reddish or coppery skin color.

* Thin, pale brown scales of exudate develop in the axillae and groin and within 3-5 days spread to all parts of the body and rapidly become dark in color and greasy in texture.

臨床症狀 (Clinical signs)

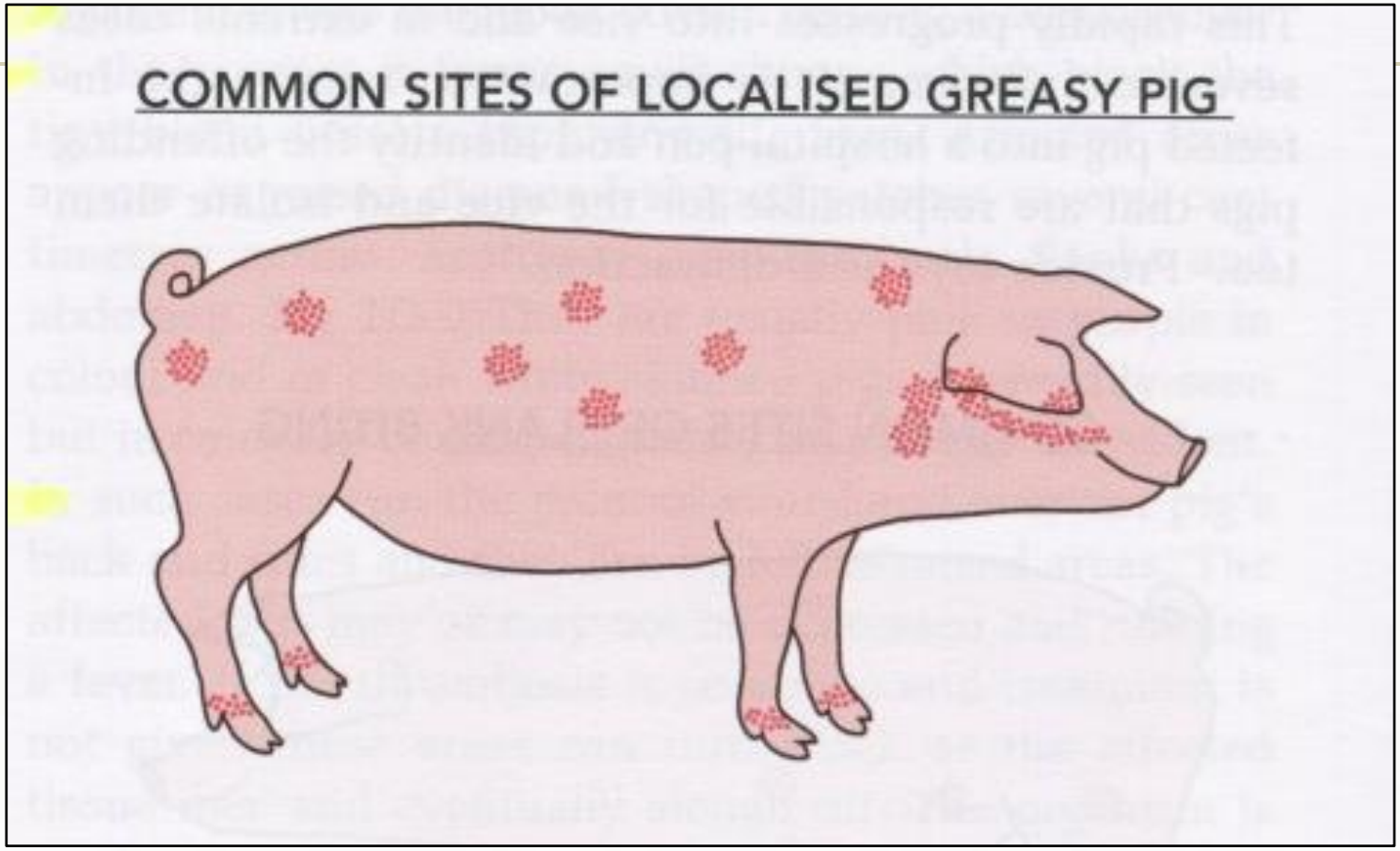
- The skin of affected piglets often feels hot, the hair coat is matted (糾結的), and the exudate may extend to the eyelashes.
- Ulcers may occur in the mouth.
- Separation of horn may occur at the bulbs of the heels.
- Overall morbidity and mortality usually are high, especially in younger pigs.

病變 (Lesions)

- Early lesions include reddening of skin and the presence of a clear exudate
- Dehydrated and odoriferous.
- The crusts are often fissured (裂縫) and fissures contain dirt and filth from the environment.
- Besides haired areas, lesions may be observed on the feet, mouth and tongue.

病變 (Lesions)

- Early changes of the epidermis:
  - exfoliation
  - exocytosis
  - crust formation
  - formation of vesicles and pustules (intraepidermal vesicular and pustular dermatitis 表皮內水泡性及膿泡性皮炎)
- Later stages
  - acanthosis (hyperplasia of the epidermis 棘層肥厚)
  - perivascular inflammation occurs in the dermis



Muirhead and Alexander, Managing Pig Health 2nd ed., 2013



The generalized form



A localized lesion of greasy pig disease. This may be cannibalized

Muirhead and Alexander, Managing Pig Health 2nd ed., 2013



Severe facial and neck lesions



Pig showing lesions on the head

Handbook of Pig Medicine
by Jackson and Cockcroft ,
2007, Elsevier



皮膚起皺褶，食慾減退，消瘦及脫水。



皮毛糾結如蠟膜包裹，皮膚龜裂，雜以皮脂、血清、穢物和皮膚碎屑，乃至體表結痂。

豬病診療 楊平政等 1996
台灣養豬科學研究所



An affected litter of piglets with both severely diseased animals and normal litter mates.



Early disease with marked inflammation of skin.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



Lesions of generalized acute exudative epidermitis. Note the moist appearance of the axillae and the color of the skin.



A subacute case with multifocal lesions.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



淡棕色皮膚屑片及滲出物出現在腋下與鼠蹊部，然後再遍步全身。患豬體表有熱感，皮毛糾結，厭食及脫水。



患豬怕冷擠成一堆取暖，皮膚起皺褶，食慾減退，消瘦及脫水。

劉振軒等著 豬病與屠宰衛生檢查圖譜
1997 台灣養豬科學研究所



四週齡哺乳豬，皮膚有油脂樣分泌物，明顯充血並有痂皮形成。



耳朵出現油脂樣滲出物，還可見多發性黃白色膿疱，潰瀾後之膿瘍常可分離到鏈球菌。

劉振軒等著 豬病與屠宰衛生檢查圖譜
1997 台灣養豬科學研究所

治療與預防 (Treatment and prevention)

- Early antimicrobial treatment is most effective. Antimicrobial selection should ideally be guided by susceptibility testing.
- Supportive care and topical disinfection may aid recovery.
- Autogenous vaccination may be considered in recurrent herd problems

治療與預防 (Treatment and prevention)

- Efforts should be made to minimize skin breaches by clipping needle teeth, removing abrasive surfaces.

- Thorough cleaning and disinfection of facilities should be carried out between farrowings, and sows entering facilities should be washed and disinfected.

- Control may depend largely on preventing trauma and improving the environment with better ventilation, cleaner and drier pens, controlled humidity, and reduced stocking density.

111年度國考試題

下列有關豬滲出性皮膚炎之敘述，何者錯誤？

A.致病原為*Staphylococcus hyicus*

B.主要發生在母豬

C.與免疫力低下有關

D.皮膚出現傷口為前置因子

109年度國考試題

下圖病豬有皮毛糾結、皮膚龜裂、體表結痂等外觀。若致病原為一種革蘭氏陽性球菌，下列敘述何者錯誤？

A. 病豬有劇烈搔癢症狀

B. 此病又稱為豬油脂病

C. 此致病原在人類不會造成疾病

D. 致病原分泌的剝落毒素（exfoliative toxin）是重要的致病因子



108年度國考試題

下列有關豬滲出性表皮炎之敘述，何者正確？

A.屬於人畜共通傳染病

B.病豬皮脂過度分泌、脫皮並產生劇癢

C.常因免疫力降低而感染

D.由感染皮黴菌所引起

105年度國考試題

下列何者為豬隻滲出性皮膚炎之病原所產生之毒素？

A. ExhA與ExhB

B. ShetC與ShetD

C. PmtA與PmtB

D. Stx2e與Stx2d

104年度國考試題

有關豬滲出性表皮炎（exudative epidermatitis）的相關敘述，下列何者正確？

A. 病原不屬於常在菌

B. 病原會引起人的皮膚病

C. ivermectin常用來治療此疾病

D. 病原為革蘭氏陽性球菌

103年度國考試題

下列何者為豬滲出性表皮炎的病因？

A. *Streptococcus*

B. *Clostridium*

C. *Pasteurella*

D. *Staphylococcus*

100年度國考試題

下列有關豬滲出性表皮炎之敘述，何者錯誤？

A.僅感染豬

B.好發於3月齡以下豬

C.多因創傷後發生

D.有劇癢症狀

99年度國考試題

* 下列何者可有效治療豬油脂病 (greasy-pig disease)?

  A. 外寄生蟲驅蟲劑

  B. 內寄生蟲驅蟲劑

  C. 抗菌劑

98年度國考試題

* 下列有關豬油脂病 (greasy-pig disease)之敘述，何者正確?

A. 細菌性疾病

B. 病毒性疾病

C. 營養性疾病

D. 腫瘤性疾病

96年度國考試題

豬油脂病（greasy-pig disease）的病原為下列何者？

A. 鏈球菌

B. 梭菌

C. 沙門氏菌

D. 葡萄球菌

96年度國考試題

豬隻滲出性表皮炎（exudative epidermitis）之病原為下列何者？

A. *Staphylococcus aureus*

B. *Staphylococcus epidermidis*

C. *Staphylococcus hyicus*

D. *Streptococcus pyogenes*

布氏桿菌症 (brucellosis)

關聯 (Relevance)

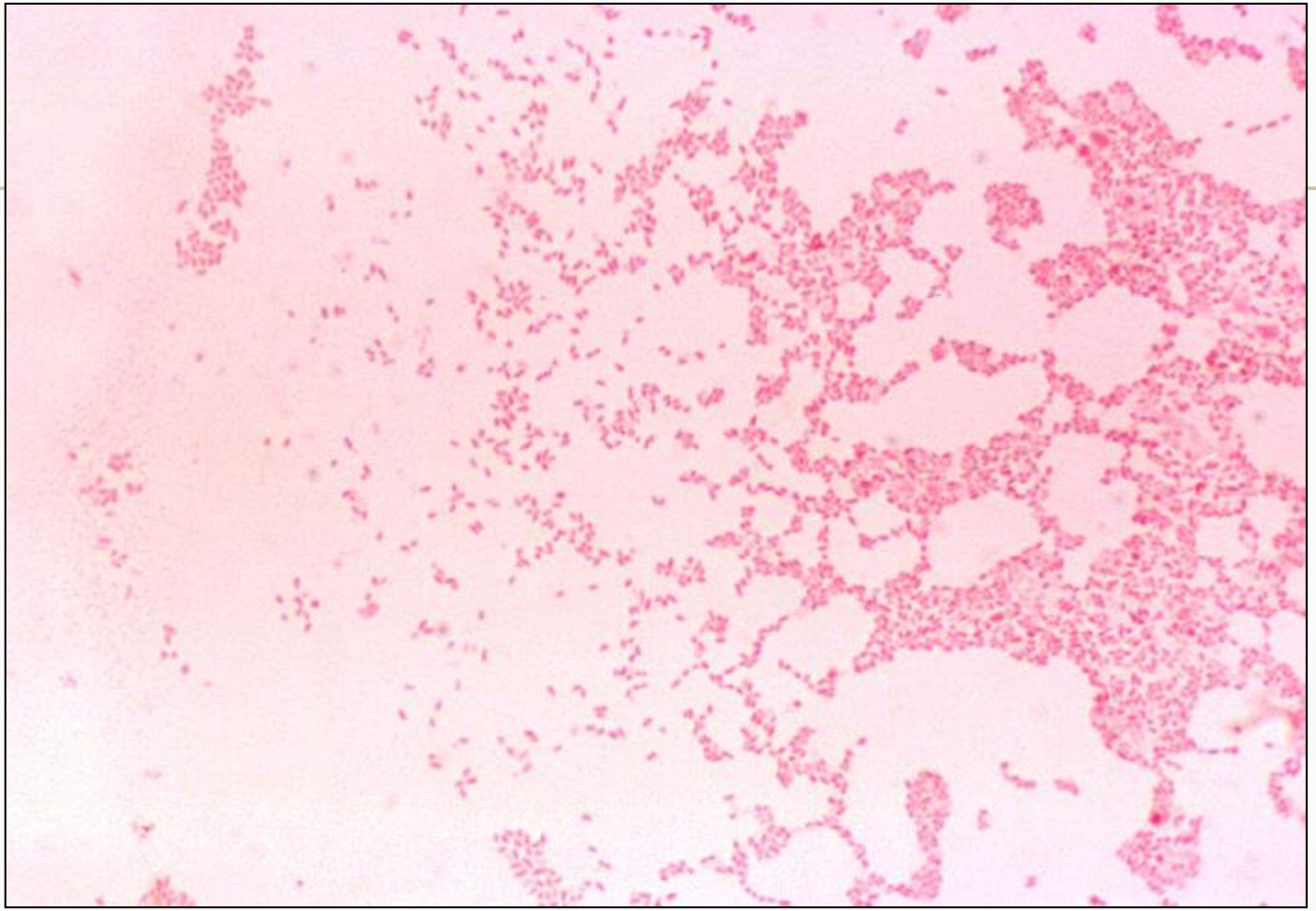
- Swine brucellosis persists in several regions worldwide, particularly where infected domestic or feral pigs are present. Feral swine can serve as important reservoirs.

- Swine brucellosis is essentially eradicated in the commercial production swine herd in the United States. This disease appears to be widespread in South America. In Europe, there is a general low prevalence of swine brucellosis.

- Asia, particularly Southeast Asia, seems to have a generally high prevalence of the disease, predominantly caused by biovar 3 in south China and Singapore and by biovar 1 elsewhere.

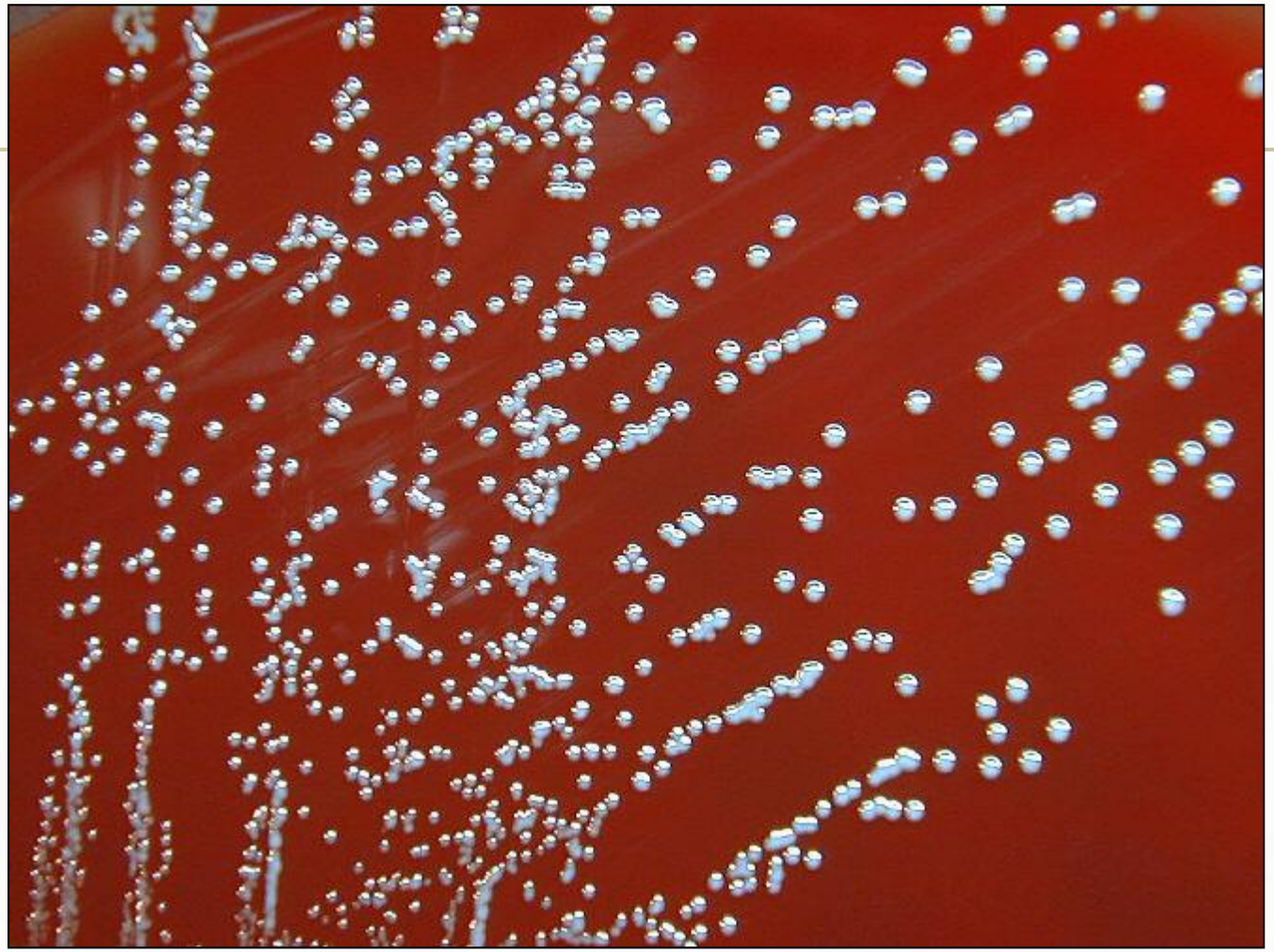
病因 (Etiology)

- *Brucella*: Gram negative, small bacilli or coccobacilli, nonmotile and arranged singly.
- *Brucella suis* is the principal *Brucella* species causing swine brucellosis and reproductive disease in pigs.
- *Brucella* is a major laboratory-acquired infection risk.



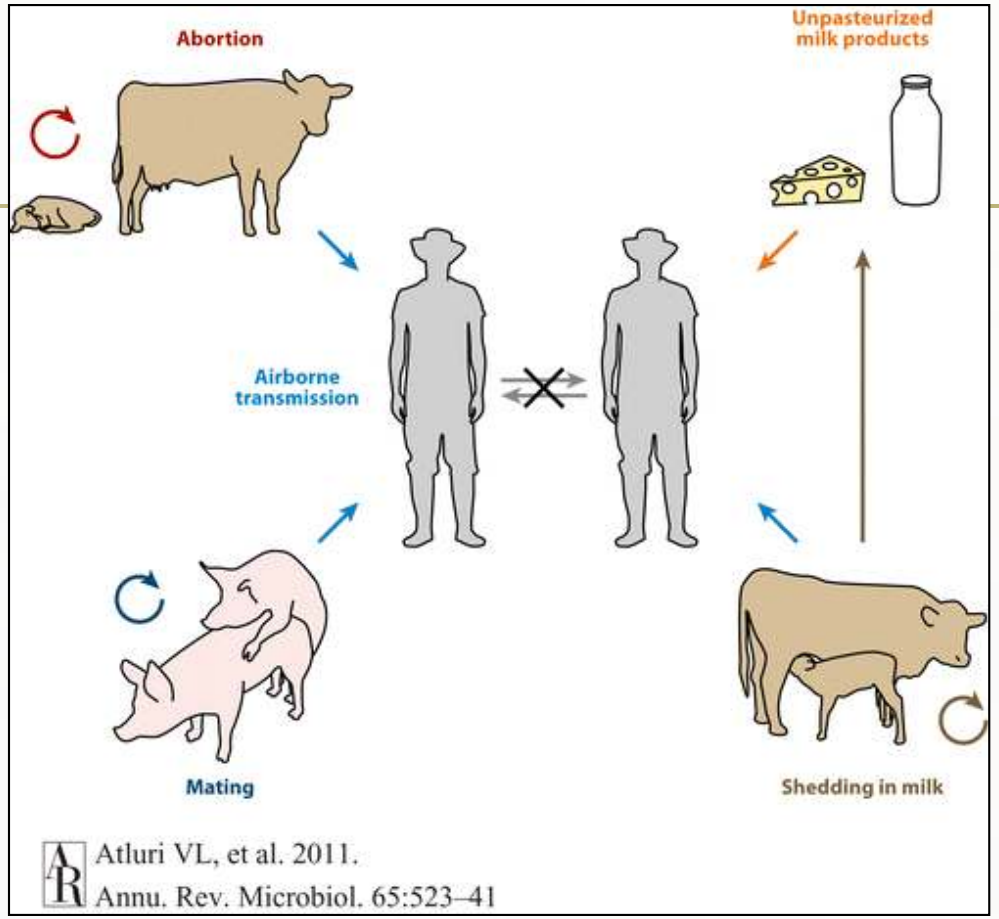
A Gram-stain of *Brucella* showing the coccobacillary shape and Gram-negative staining

CDC, public health image library 1937



Brucella grows slowly, even on rich media to give pinpoint, translucent colonies with a smooth surface

CDC, public health image library 1902



Ecology of zoonotic *Brucella* infections. Depending on the *Brucella* species involved, transmission within the natural host occurs via abortion, via genital contact during mating, or via shedding in milk. Zoonotic transmission occurs most frequently via unpasteurized milk products in urban settings, while occupational exposure of farmers, veterinarians, or laboratory workers can result from aerosols containing *Brucella* spp. Only rare cases of human-to-human transmission have been reported.

Annu. Rev. Microbiol. 65: 523-541, 2011

Host preference and zoonotic potential of *Brucella* species

Species	Host preference	Relative importance in human infection
Brucella melitensis	Sheep, goat	high
Brucella abortus	Cattle	moderate
Brucella suis	Pig	moderate
Brucella canis	Dog	mild
Brucella cetis	Dolphin, porpoise, whale	mild
Brucella pinnipedialis	Seal	mild
Brucella inopinata	Unknown	mild
Brucella ovis	Sheep	No reported
Brucella neotomae	Desert woodrat	No reported
Brucella microti	Common vole	No reported
Brucella sp. (baboon isolate)	Baboon	No reported

Annu. Rev. Microbiol. 65: 523-541, 2011

公共衛生 (Public health)

- *B. melitensis*, followed by *B. abortus*, and *B. suis* are the main species involved in the infection of human beings.
- Brucellosis remains an important zoonosis worldwide, especially in regions where infection persists in livestock or wildlife reservoirs.
- In North America, Australia, and some northern European countries, control of *Brucella* in zoonotic reservoir species by vaccinating and culling of infected animals has strongly reduced the incidence of human brucellosis.

公共衛生 (Public health)

- Porcine brucellosis also has noteworthy public health implications. Until recently the source of the majority of human brucellosis has been *Brucella suis*-infected pigs.

- *B. suis* (biovar 1 and biovar 3) appears to have a much higher degree of pathogenicity for humans than *B. abortus*.

- Incidence of *B. suis* in humans is almost entirely occupational: farmers, veterinarians, abattoir workers, hunters etc.

流行病學 (Epidemiology)

- Most *B. suis* infection is transmitted to susceptible animals through direct association with infected pigs.

  --alimentary tract: eat food or drink fluid contaminated with discharges from infected pigs.

  --mating: sows are infected when mated with infected boars or when artificially inseminated with semen containing *B. suis*.

流行病學 (Epidemiology)

- Brucellae can survive in organic matter at freezing or near-freezing temperature. Efforts to eliminate brucellosis from pig raising premises must include an effective sanitation program.

- The epidemiological importance of feral pigs in the maintenance of porcine brucellosis depends largely on the degree of contact between wild and domestic pigs.

- Pigs infected with *B. suis* biovar 1, 2, or 3 can serve as a source of infection for other domesticated animal species. *B. suis* infection can occur naturally in horses, cattle, and dogs.

致病機制 (Pathogenesis)

Entry into the host

Urease likely serves to buffer the low pH of the stomach by producing CO_2 from urea.

Persistence at systemic sites

Brucella has relatively low-endotoxic LPS and multiple mechanisms that help evade innate immune recognition and complement-mediated killing.

致病機制 (Pathogenesis)

Intracellular survival

Acidic environment within the vacuole provides a signal for induction of the VirB type IV secretion system, a virulence factor that is crucial for intracellular replication of *Brucella* in macrophages or dendritic cells.

致病機制 (Pathogenesis)

- Erythritol (古柯二醇) and other host factors may favor *Brucella* growth in reproductive tissues, contributing to genital tropism.

- Animal tissues, including breast, uterus, placenta, and epididymis, are rich in erythritol.

臨床症狀 (Clinical signs)

- The classic manifestation of pig brucellosis are:

  --abortion

  --infertility

  --orchitis (睾丸炎)

  --posterior paralysis and lameness

- Genital infection tends to be more persistent in boars than in sows. Boars that do develop genital infection seldom recover from it. Infection in accessory genital glands would disseminate large numbers of *B. suis* in the semen.

臨床症狀 (Clinical signs)

- Pathological changes in the male accessory glands or testes are generally more extensive and irreversible than in the uterus. Infertility and lack of sexual drive may occur in the infected boars and are frequently associated with testicular involvement.

- The rate of abortion is highest in sows or gilts exposed via the genital tract at the time of breeding.

- The majority of female pigs eventually recover from genital infection.

病變 (Lesions)

- Generally, the histopathological changes consist of uterine glands filled with leukocytes, cellular infiltration of the endometrial stroma, and hyperplasia of periglandular connective tissue.

- Diffuse suppurative inflammation is usually present in affected placentas, resulting in purulent, necrotizing placentitis with fibrinous polyserositis.

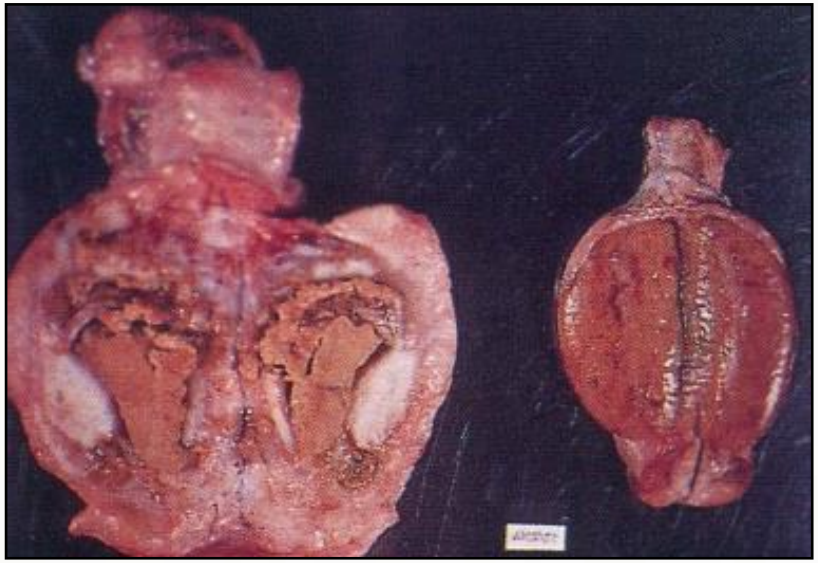
- Focal microscopic granulomatous lesions frequently can be observed in livers of pigs with brucellosis.

病變 (Lesions)

- Microscopic lesions of bones are sometimes caused by *B. suis* infections. The lesions are most frequently located adjacent to the epiphyseal cartilage and usually consist of caseous centers surrounded by a zone of macrophages and leukocytes and often by an outer zone of fibrous connective tissue.



睾丸腫大

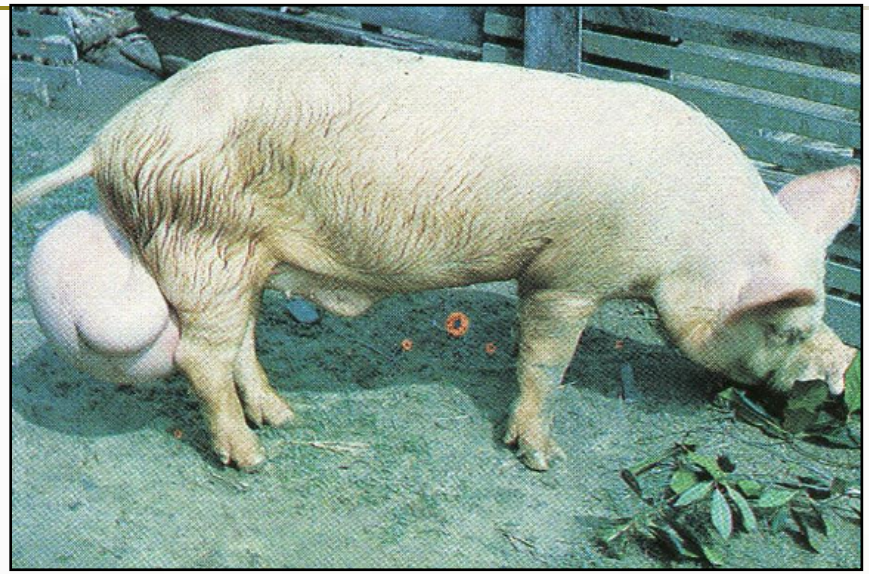


貯精嚢炎、腫大、化膿

豬病診療 楊平政等 1996
台灣養豬科學研究所

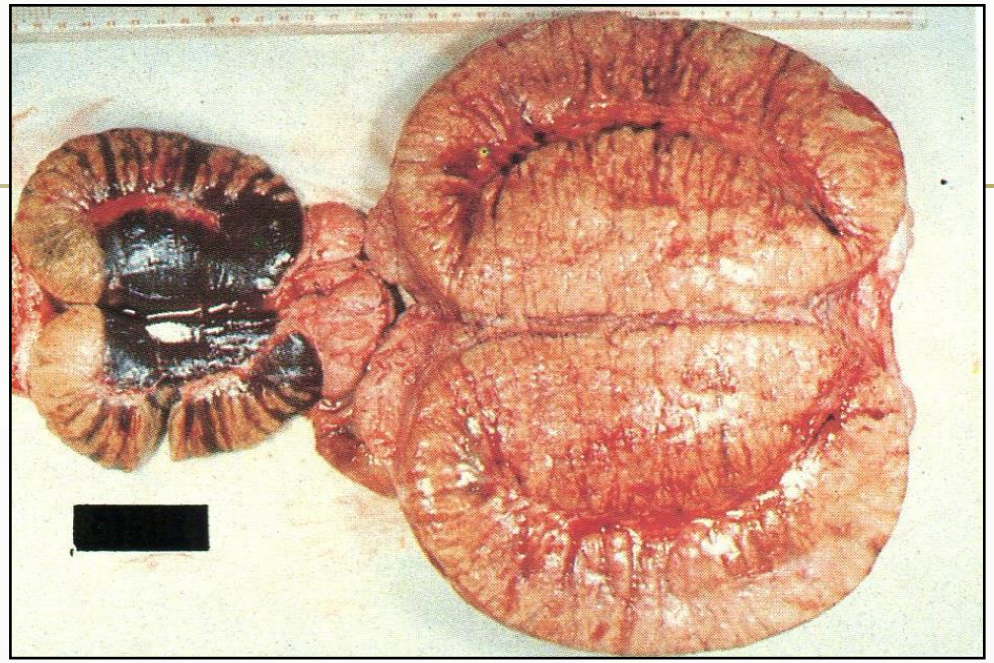


This boar has enlargement of the left testicle caused by infection with *B. suis*.



A large white boar with *B. suis* orchitis. Note the enlargement of the testis.

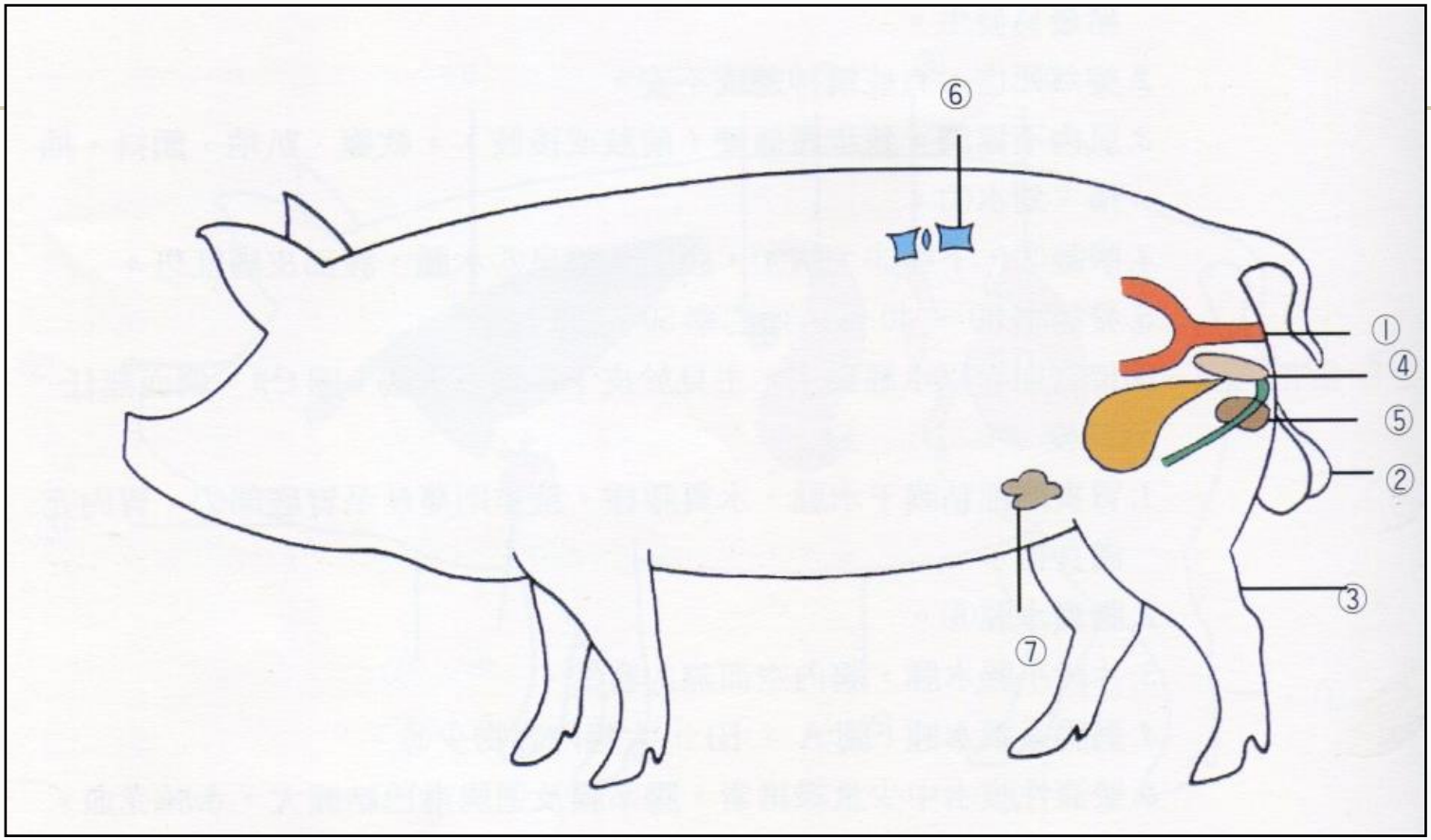
Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.



Longitudinal section of porcine testes illustrating degrees of orchitis due to *B. suis* infection. The gross enlargement of testis on the right is due to necrosis of the seminiferous tubules with thickening of the interstitial tissue. The smaller testicle on the left contains pale areas of necrotic seminiferous tubules.

Smith et al., *a colour atlas of diseases & disorders of the pig*, 1990, Wolfe Publishing Ltd.

Brucellosis



6: 椎間板壞死膿腫及鄰近脊椎骨炎
7: 淋巴結發炎壞死常見肉芽腫性淋巴炎

豬病診療 楊平政等 1996
台灣養豬科學研究所



The skeleton (including joints and intervertebral discs as shown here) is often infected by *Brucella suis* in pigs.

http://www.kosvi.com/courses/vpat5215_1/vpat5310/fetus/fet07.htm

診斷 (Diagnosis)

- Serology is commonly used for herd screening.
- Culture provides definitive identification but poses substantial biosafety risk.

治療與控制 (Treatment and control)

- Safe and reliable vaccines that produce serviceable immunity against brucellosis in pigs have not been developed.

- Control of infected commercial herds relies primarily on testing, removal/culling of infected animals, strict biosecurity, and, in some eradication programs, herd depopulation.

治療與控制 (Treatment and control)

- Antimicrobial treatment is generally not recommended as a herd-eradication strategy.
- *Brucella* can persist intracellularly and relapse after therapy.
- Treatment does not reliably eliminate infection or zoonotic risk.
- Test-and-remove/culling and biosecurity are preferred control approaches.

107年度國考試題

有關豬布氏桿菌症（brucellosis）之敘述，下列何者錯誤？

A. 目前沒有可靠及安全的疫苗可使用

B. 可能會有椎間板壞死膿腫及鄰近脊椎骨炎

C. 肝臟可能會有肉芽腫病灶

D. 公豬的臨床症狀通常較母豬容易復原

106年度國考試題

下列有關布氏桿菌與其主要感染動物的配對中，何者錯誤？

A. *Brucella canis*：犬

B. *Brucella melitensis*：羊

C. *Brucella abortus*：牛

D. *Brucella ovis*：豬

103年度國考試題

豬隻的胎盤、乳腺、副睪等組織含有較高濃度的erythritol成分，有利於下列那一種細菌的繁殖?

A. *Brucella*

B. *Salmonella*

C. *Streptococcus*

D. *Yersinia*

101年度國考試題

下列何者不是布氏桿菌病之病原？

A. *Brucella fetus*

B. *Brucella suis*

C. *Brucella canis*

D. *Brucella melitensis*

99年度國考試題

* 豬布氏桿菌症最主要的病原為下列何者？

  A. *Brucella melitensis*

  B. *Brucella ovis*

  C. *Brucella abortus*

  D. *Brucella suis*

Take Home Message

1. *E. rhusiopathiae* → acute septicemia, diamond skin, chronic arthritis/endocarditis.
2. Penicillin is the classic first-line treatment for acute erysipelas.
3. *S*. Choleraesuis → systemic/septicemic salmonellosis.
4. *S*. Typhimurium → enterocolitis.
5. Carrier pigs are central to *Salmonella* epidemiology and food safety.
6. *S. hyicus* → exfoliative toxin → greasy pig disease.
7. Exudative epidermitis is usually nonpruritic.
8. *B. suis* → abortion, infertility and orchitis.
9. Brucellosis is an important zoonotic and laboratory hazard.
10. Brucellosis control relies on eradication/biosecurity rather than treatment.